INCH-POUND
MIL-M-28787/370
26 October 1990

MILITARY SPECIFICATION

MODULE, STANDARD ELECTRONIC

NETWORK, SPECIAL, KEYCODE HJL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of a special electronic network module. This specification covers one class of module, class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.0786 failures per million operating hours for class I, when predicted in accordance with MIL-STD-1389.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be HJL.

3.6.2 Size. The module size shall be 1B Format B.

3.6.3 Weight. The module shall have a maximum weight of 60 grams.

3.6.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.6.5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A, and a heat dissipation fin on increment B.

3.6.6 Depth. The module shall have no additional depth requirements

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/370-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/ -3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation,									
	initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A5	Isolation,									
	initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B	Maximum stress	III	Q	----- Module shall pass all remaining tests -----						-----
C1-C9	Output short circuit current	IV	S	10	40	10	40	10	40	mA
D1,D2	Continuity	V	X	0 0	1.0	0 0	1.0	0.0	1.0	Ohm
D3,D4	Continuity	V	Q	0 0	0.1	0 0	0.1	0.0	0.1	Ohm
E1,E2	Power supply current	VI	X	1	24	1	24	1	24	mA
F1,F2	PSFC	VII	X	2.07	2.53	2.07	2.53	2.07	2.53	uF
F3,F4	PSFC	VII	X	0 36	0.44	0.36	0 44	0.36	0.44	uF
G	Diode voltage drop	VIII	X	0.4	0.8	0.4	0.9	0.3	0.7	V
H1,H2	Inductor check	IX	X	35 0	80 0	35 0	80.0	35.0	80.0	Ohm
H3,H4	Inductor check	IX	Q	3 5	4.3	3.4	4.4	3.4	4.4	mH
J1-J6	Resistance check,									
	initial	X	X	37 03	37.77	37.00	37.80	36.99	37.80	Ohm
	end-of-life	X	Q	36.28	38.52	36.25	38.54	36.24	38.55	Ohm
J7-J10	Resistance check,									
	initial	X	X	74 93	75.08	74.87	75.13	74.85	75.15	kOhm
	end-of-life	X	Q	73.42	76.58	73.37	76.63	73.35	76.65	kOhm
J11,J12	Resistance check,									
	initial	X	X	529.47	530.53	528 67	531.33	528.41	531.59	kOhm
	end-of-life	X	Q	518.87	541.13	518.08	541.93	517.81	542.19	kOhm

- See notes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ -3/	25 ± 5°C 4/		0 ± 5°C 4/		6∂ ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
K1,K3,K5	IAMP offset voltage, initial	XI	X	-110	110	-160	160	-180	180	uV
	end-of-life	XI	Q	-450	450	-650	650	-750	750	uV
K2,K4,K6	IAMP offset voltage, initial	XI	X	-3 25	3.25	-4 5	4.5	-5.0	5.0	mV
	end-of-life	XI	Q	-10 0	10.0	-15 0	15.0	-18.0	18.0	mV
L1,L2,L4 L5,L7,L8	IAMP bias current, initial	XII	X	-50.5	50.5	-53.0	53.0	-54 0	54.0	nA
	end-of-life	XII	Q	-85.0	85.0	-90 0	90.0	-93 0	93.0	nA
L3,L6,L9	IAMP offset current, initial	XII	X	-35.5	35.5	-38 0	38.0	-39.0	39.0	nA
	end-of-life	XII	Q	-53.0	53.0	-57 0	57.0	-59.0	59.0	nA
M1-M9	IAMP input resistance	XIII	Q	10	INF	10	INF	10	INF	MOhm
N1-N3	IAMP gain accuracy	XIV	X	0 9985	1.0015	0 9984	1.0015	0.9984	1.0015	V/V
N4-N6	IAMP gain accuracy	XIV	X	9.975	10.025	9.974	10.026	9.974	10.026	V/V
N7-N9	IAMP gain accuracy	XIV	X	99.5	100 5	99.5	100.5	99.4	100.6	V/V
N10-N12	IAMP gain accuracy	XIV	X	980 0	1020 0	977.0	1023.0	976.0	1024.0	V/V
P1-P3	Output voltage swing	XV	X	9.5	11 0	9.5	11.0	9.5	11.0	V
P4-P6	Output voltage swing	XV	X	-11.0	-9 5	-11.0	-9.5	-11 0	-9.5	V
Q1-Q3	IAMP CMRR	XVI	S	70	INF	70	INF	70	INF	dB
Q4-Q6	IAMP CMRR	XVI	S	90	INF	90	INF	90	INF	dB
Q7-Q9	IAMP CMRR	XVI	S	100	INF	100	INF	100	INF	dB
Q10-Q12	IAMP CMRR	XVI	S	110	INF	110	INF	110	INF	dB
R1-R3	IAMP PSS	XVII	S	75	INF	75	INF	75	INF	dB
R4-R6	IAMP PSS	XVII	S	95	INF	95	INF	95	INF	dB
R7-R9	IAMP PSS	XVII	S	105	INF	105	INF	105	INF	dB
R10-R12	IAMP PSS	XVII	S	110	INF	110	INF	110	INF	dB

- See notes at end of table

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TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ -3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
S1-S3	IAMP settling time	XVIII	Q	0	90	0	90	0	90	uS
S4-S9	IAMP settling time	XVIII	Q	0	30	0	30	0	30	uS
S10-S12	IAMP settling time	XVIII	Q	0	150	0	150	0	150	uS
T1-T9	IAMP slew rate	XIX	X	3.0	INF	3.0	INF	3.0	INF	V/uS
T10-T12	IAMP slew rate	XIX	X	0.5	INF	0.5	INF	0 5	INF	V/uS
U1-U3	BAMP offset voltage,									
	initial	XX	X	-2.0	2.0	-6.0	6.0	-6.0	6.0	mV
	end-of-life	XX	Q	-3.0	3 0	-10.0	10.0	-10.0	10.0	mV
V1-V3	BAMP bias current,									
	initial	XXI	X	-10.0	10 0	-14.0	14.0	-14.0	14.0	nA
	end-of-life	XXI	Q	-15.0	15 0	-21.0	21.0	-21.0	21.0	nA
W1-W3	BAMP voltage swing	XXII	X	9.0	11 0	9.0	11 0	9.0	11 0	V
X1-X3	BAMP slew rate	XXIII	X	3 75	INF	3.75	INF	3.75	INF	V/uS
Y1-Y3	Output voltage swing	XXIV	X	9.5	11 0	9.5	11.0	9.5	11.0	V
Z1-Z3	RMS-DC offset voltage,									
	initial	XXV	X	-1.5	1.5	-3 0	3.0	-3 0	3.0	mV
	end-of-life	XXV	Q	-2.5	2.5	-4 5	4 5	-4.5	4.5	mV
AA1-AA3	RMS-DC input resistance	XXVI	X	6.4	9 6	6.4	9.6	6.4	9.6	kOhm
BB1-BB6	RMS-DC nonlinearity	XXVII	X	-0 1	0.1	-0.2	0 2	-0.3	0.3	percent
CC1-CC12	RMS-DC PSS	XXVIII	Q	0	300	0	300	0	300	uV/V
DD1-DD3	RMS-DC reversal error	XXIX	X	-0 25	0 25	-0 25	0 25	-0 25	0.25	percent
EE1-EE3	RMS-DC total error	XXX	X	6.964	7.036	6 964	7 036	6.964	7.036	V

- See notes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ -3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
FF1-FF3	Crest factor error	XXXI	X	-5 0	5.0	-5 0	5.0	-5 0	5.0	percent
GG1-GG3	Ripple voltage	XXXII	Q	54	115	54	115	54	115	mVrms
HH1-HH3	RMS-DC input resistance	XXXIII	Q	20	30	20	30	20	30	kOhm
JJ1-JJ9	Frequency response	XXXIV	Q	-4 0	4 0	-4 0	4.0	-4 0	4.0	percent

NOTES

1/ 'X' parameters shall be 100 percent tested during module acceptance test.

2/ 'S' parameters shall be sample acceptance tested (1 percent or 2 modules from each lot whichever is greater) at 25°C

3/ 'Q' parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used are defined in 6.1

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3.7 Circuit pin configuration. The functional block diagram and pin assignments shall conform to figure 1.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25 degrees C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for power supply voltages +15 V dc ± 10 percent and -15 V dc ± 10 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

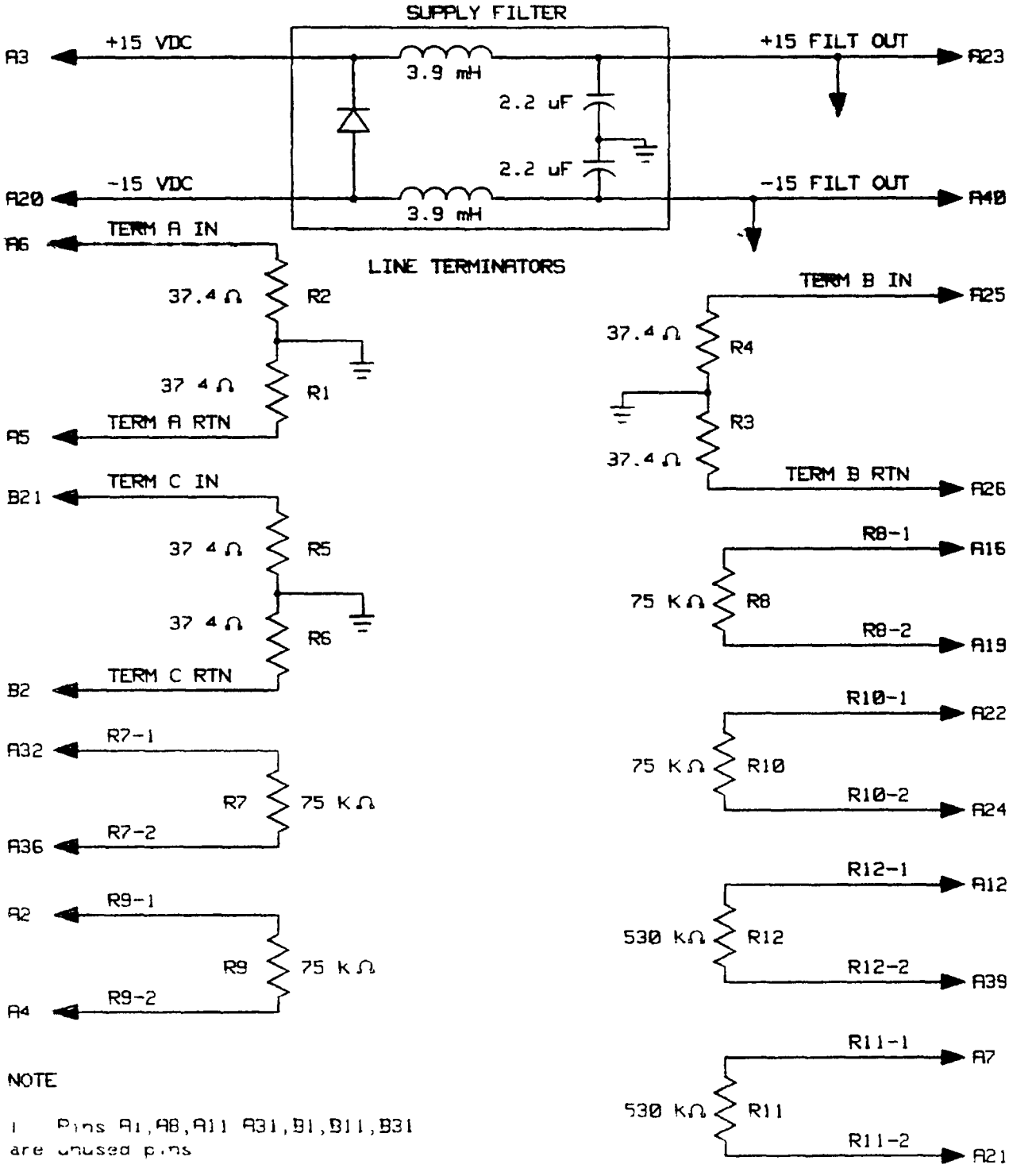
4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4 4.2 Order of testing. Test conditions A through JJ shall be performed in any order with the following exceptions. Test conditions A through C are stress tests and shall be performed only at 25 ± 5 degrees C. If test conditions A through D are part of a sequence of tests, the order of testing shall be as follows: A, B, C and D where applicable, followed by the remaining tests in the sequence

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NOTE

1. Pins A1, A8, A11, A31, B1, B11, B31 are unused pins.

FIGURE 1. HJL functional block diagram and pin assignments

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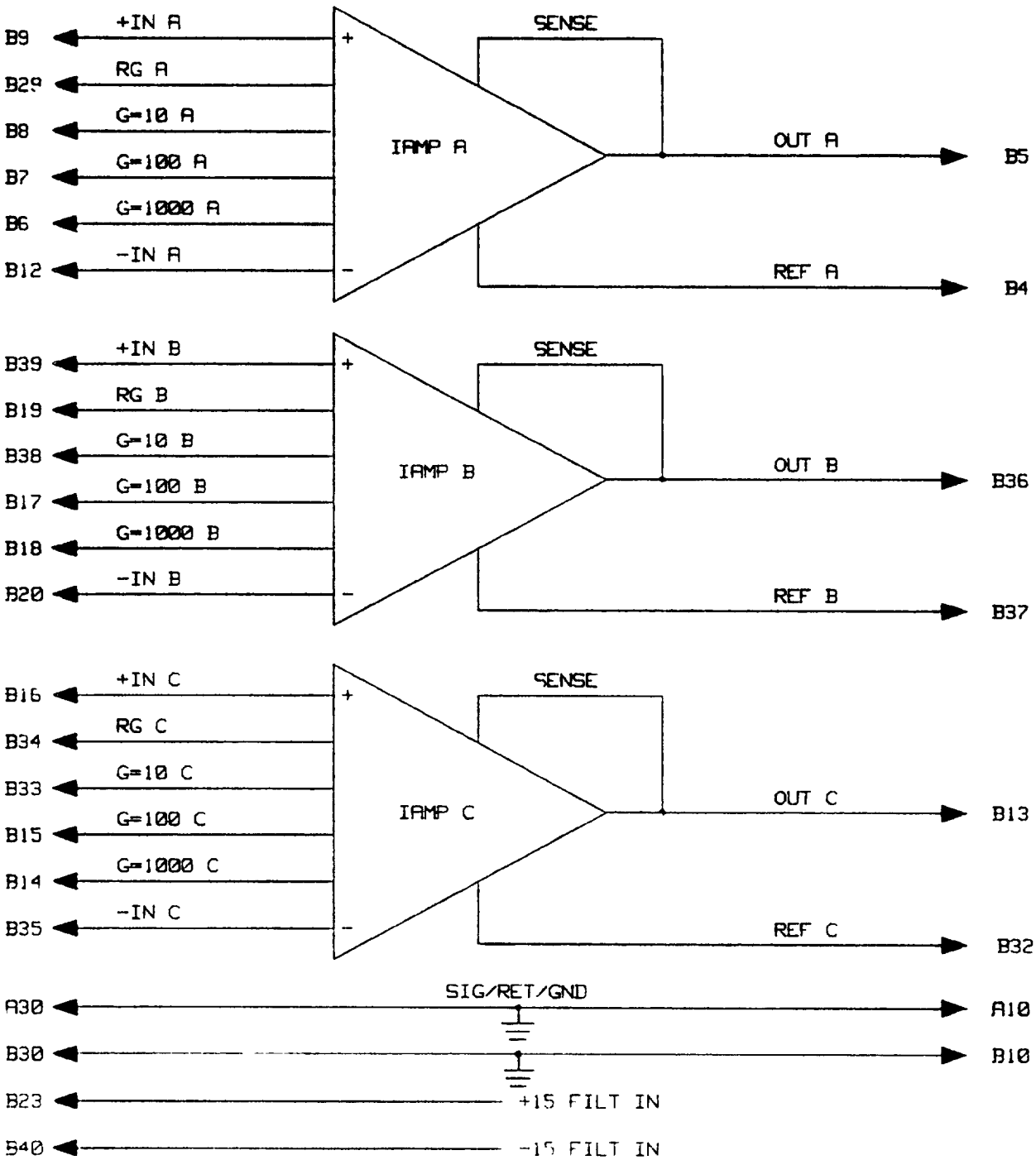


FIGURE 1 HJL functional block diagram and pin assignments - Continued

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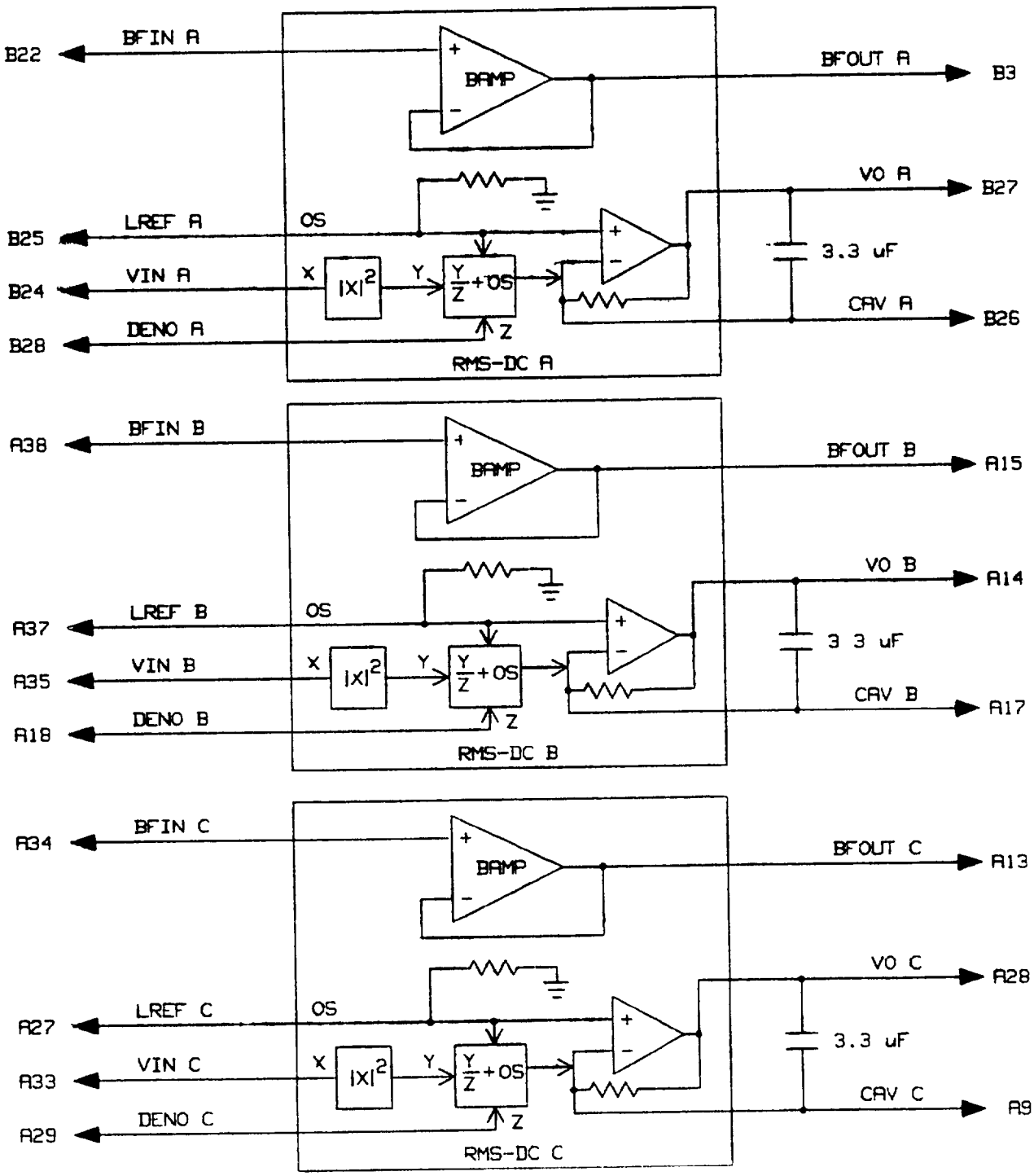


FIGURE 1 HJL functional block diagram and pin assignments - Continued

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4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through JJ in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25 degrees C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through JJ shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

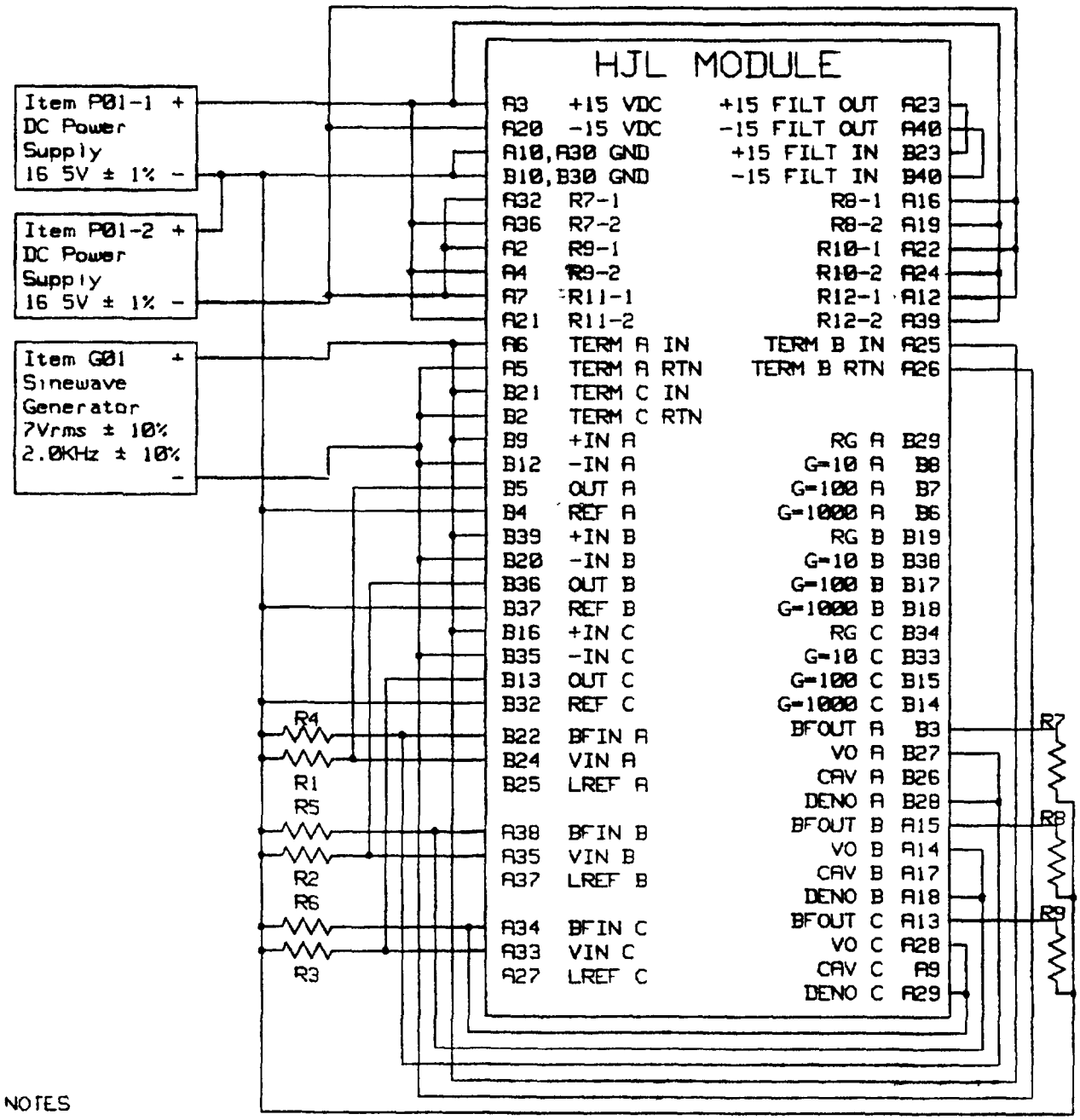
4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition F, shall be performed using the initial limits at 25 ± 5 degrees C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests D through JJ shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5 degrees C, electrical tests D through JJ shall be performed in the order specified in 4 4.2 using the 25 ± 5 degrees C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5 degrees C ambient.

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NOTES

1. R1-R3 = 1 5 kOhm ± 1.0 percent, 0.5 W

2 R4-R9 = 2 0 kOhm ± 1 0 percent, 0 5 W

FIGURE 2 HJL operating temperature, component temperature and life test circuit

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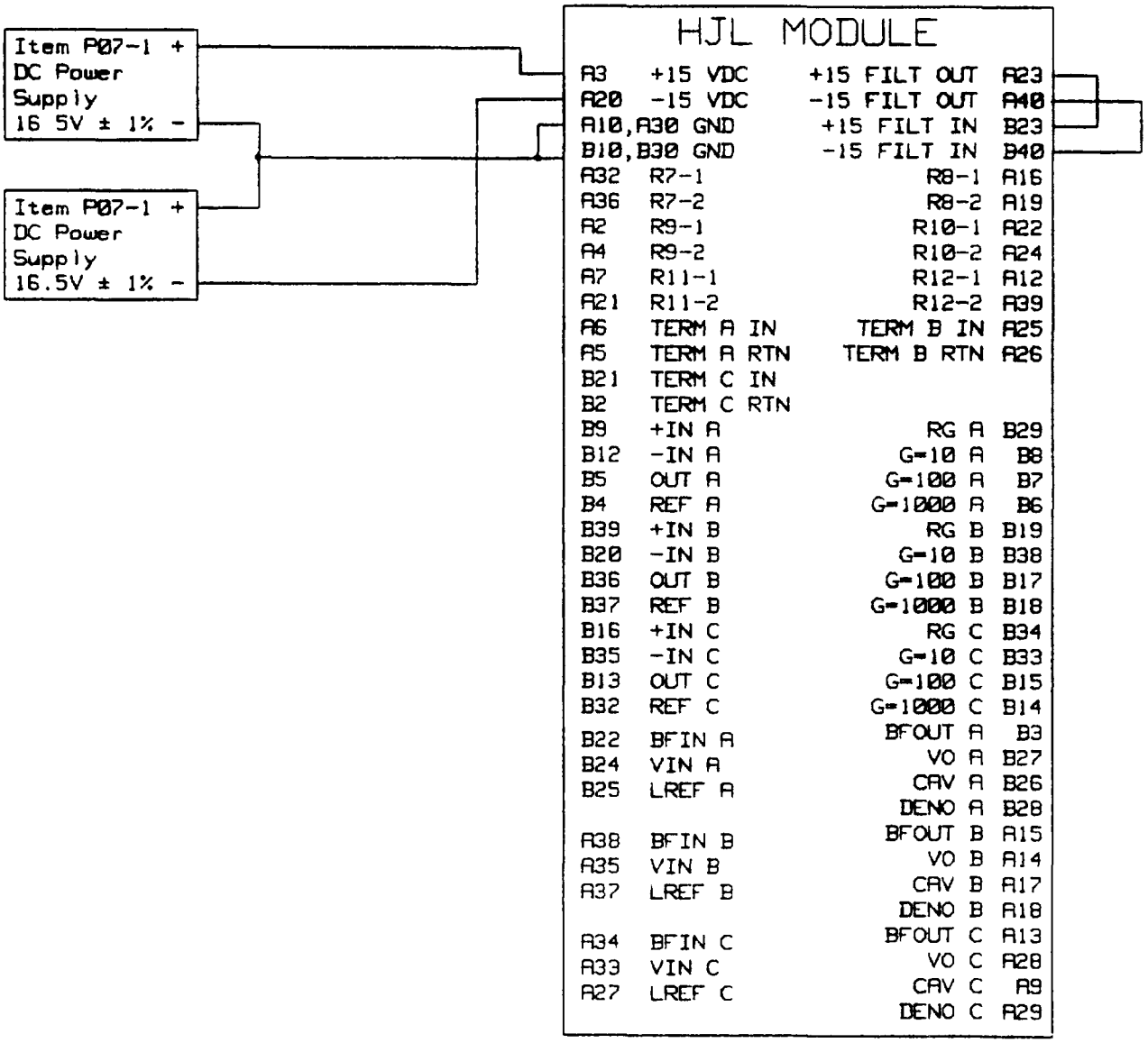


FIGURE 3 HJL vibration test circuit

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions.

(a) PSFC = power supply filter capacitance.
(b) PSS = power supply sensitivity.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0 degrees C to 60 degrees C end-of-life parameters in table I.

Power dissipation	1.2 W max.
Supply voltages	+15 V dc ± 10 percent -15 V dc ± 10 percent
Absolute maximum voltage supply ratings	±16.5 V dc
Power supply capacitance	2.3 uF ± 10 percent
Power supply inductance	4.4 mH ± 23 percent
Analog input characteristics	
Input voltage ranges	
IAMP	±10 V
RMS-DC converter	±10 V
BAMP	±12.5 V
Input impedance	
IAMP	100 MOhm
RMS-DC converter	8 kOhm ± 20 percent
BAMP	100 MOhm

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6.4 Module function. This special network module is a 1B size standard electronic module that includes three individual Instrumentation Amplifiers (IAMPs), three individual Root-Mean-Square to Direct Current (RMS-DC) converters, three Buffer Amplifiers (BAMPs), three individual center-tapped 75 Ohm line terminators together with resistor networks and power supply line filters. The three IAMPs improve common mode rejection. The optional non-inverting unity gain BAMPs are included to increase the module's output current drive capability to five milliamperes.

6.4.1 Instrumentation amplifiers. Each IAMP has pin programmable fixed gains of 1, 10, 100, and 1000. It has high linearity, high common mode rejection, low offset voltage drift, and low noise. It has an output slew rate of 3 volts/usecond and settles to within 0.01 percent in 30 useconds when programmed for a gain of 100 and 150 useconds with a gain of 1000. Although the IAMP is desensitized to common mode voltage due to a high common mode rejection ratio (CMRR), the maximum common mode voltage is restricted such that none of the stages internal to the IAMP saturate. The maximum allowable common mode voltage is Vcm=12-(A/2*Vd) where A is the amplifier gain and Vd is the difference mode voltage.

6.4.2 RMS-DC converter. The RMS-DC converter is a monolithic integrated circuit that is capable of converting the true RMS value of a complex (ac plus dc) waveform with a crest factor up to seven with less than one percent additional error. The input circuitry of the RMS-DC converter is protected from overload voltages that are in excess of supply levels. A reference pin is provided to trim the output offset.

6.4.3 Line terminators The three 75 Ohm line terminators each consist of two 37.4 Ohm, one third watt resistors in series with the center-tap tied to analog ground.

6.4.4 Resistor network The resistor network consists of four individual 75 kOhm resistors and two 530 kOhm, 0.01 watt resistors for ground fault detection. Application may also include current limiting.

6.4.5 Power supply filters. Inductance capacitance filters are provided for all 15 volt supply inputs ensuring a minimum power supply rejection ratio of 50 dB. This also allows operation of the HJL module with switching type 15 volt supplies. A diode provides suppression of current arcing at the power input pins in case of hot socket extraction.

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6.4.6 Functional description of I/O pins.

Signal name	Pin no.	Description
+15 FILT IN	B23	Filtered +15 volt supply internal to the module.
-15 FILT IN	B40	Filtered -15 volt supply internal to the module.
+15 FILT OUT	A23	Filtered output of the +15 volt supply normally connected to +15 FILT IN.
-15 FILT OUT	A40	Filtered output of the -15 volt supply normally connected to -15 FILT IN.
+15 VDC	A3	Positive 15 volt supply to module.
-15 VDC	A20	Negative 15 volt supply to module.
SIG/RET/GND	A10 A30 B10 B30	Power supply and circuit ground. Grounds from both circuit boards should be connected together.
VIN A VIN B VIN C	B24 A35 A33	Input to the RMS-DC converter.
LREF A LREF B LREF C	B25 A37 A27	External voltage reference input to the RMS-DC converter.
DENO A DENO B DENO C	B28 A18 A29	Denominator input to the RMS-DC converter. The output of the converter is normally connected to the denominator input for true RMS conversion.
VO A VO B VO C	B27 A14 A28	The true RMS output of the RMS-DC converter

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6.4.6 Functional description of I/O pins - Continued.

Signal name	Pin no.	Description
CAV A CAV B CAV C	B26 A17 A9	Averaging time constant input. A 3.3 microfarad capacitor is connected in parallel with a 24 kilohm resistor. This pin is used only to change the time constant.
BFIN A BFIN B BFIN C	B22 A38 A34	This is the input to the non-inverting unity gain BAMP. It is solely used to buffer the output of the RMS-DC converter.
BFOUT A BFOUT B BFOUT C	B3 A15 A13	This is the output from the non-inverting unity gain BAMP. It is solely used to buffer the output of the RMS-DC converter.
RG A RG B RG C	B29 B19 B34	Gain control terminal, when connected to the preset gain terminals, G10, G100, G1000.
G=10 A G=10 B G=10 C G=100 A G=100 B G=100 C G=1000 A G=1000 B G=1000 C	B8 B38 B33 B7 B17 B15 B6 B18 B14	Preset programmable gain terminals, connect to RG 2 for the desired gain. Gain is unity with no programming connection.
+IN A +IN B +IN C	B9 B39 B16	Non-inverting differential IAMP input
-IN A -IN B -IN C	B12 B20 B35	Inverting differential IAMP input.

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6.4.6 Functional description of I/O pins - Continued.

Signal name	Pin no.	Description
REF A	B4	Reference terminal, normally connected to analog ground, but can provide a means for injecting a precise offset.
REF B	B37	
REF C	B32	
OUT A	B5	IAMP output.
OUT B	B36	
OUT C	B13	
TERM A IN	A6	These 37.4 Ohm center-tapped termination resistors are connected to both ends of a differentially driven line.
TERM B IN	A25	
TERM C IN	B21	
TERM A RTN	A5	
TERM B RTN	A26	
TERM C RTN	B2	
R7-1	A32	These 75 kOhm resistors are intended for ground fault detection and current limiting applications.
R7-2	A36	
R8-1	A16	
R8-2	A19	
R9-1	A2	
R9-2	A4	
R10-1	A22	
R10-2	A24	
R11-1	A7	These 530 kOhm resistors are intended for ground fault detection applications.
R11-2	A21	
R12-1	A12	
R12-2	A39	

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II, figure 4.

(b) Maximum stress test - table III, figure 5.

(c) Output short circuit current test - table IV, figure 6.

(d) Continuity test - table V, figure 7.

(e) Power supply current test - table VI, figure 8.

(f) PSFC test - table VII, figure 9.

(g) Diode voltage drop test - table VIII, figure 9.

(h) Inductor check test - table IX, figure 10.

(i) Resistance check test - table X, figure 10.

(j) IAMP offset voltage test - table XI, figure 11.

(k) IAMP bias and offset current test - table XII, figure 12.

(l) IAMP input resistance test - table XIII, figure 13.

(m) IAMP gain accuracy test - table XIV, figure 14.

(n) Output voltage swing test - table XV, figure 15.

(o) IAMP CMRR test - table XVI, figure 16.

(p) IAMP PSS test - table XVII, figure 17.

(q) IAMP settling time test - table XVIII, figures 18 and 19.

(r) IAMP slew rate test - table XIX, figures 20 and 21.

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(s) BAMP offset voltage test - table XX, figure 22.

(t) BAMP bias current test - table XXI, figure 22.

(u) BAMP voltage swing test - table XXII, figure 23.

(v) BAMP slew rate test - table XXIII, figures 24 and 25.

(w) Output voltage swing test - table XXIV, figure 26.

(x) RMS-DC offset voltage test - table XXV, figure 27.

(y) RMS-DC input resistance test - table XXVI, figure 28.

(z) RMS-DC nonlinearity test - table XXVII, figure 29.

(aa) RMS-DC PSS test - table XXVIII, figure 29.

(bb) RMS-DC reversal error test - table XXIX, figure 29.

(cc) RMS-DC total error test - table XXX, figure 30.

(dd) Crest factor error test - table XXXI, figures 31 and 32.

(ee) Ripple voltage test - table XXXII, figure 33.

(ff) RMS-DC input resistance test - table XXXIII, figure 34

(gg) Frequency response test - table XXXIV, figure 35.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0063)

Agent:
NW

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TABLE II. Isolation test

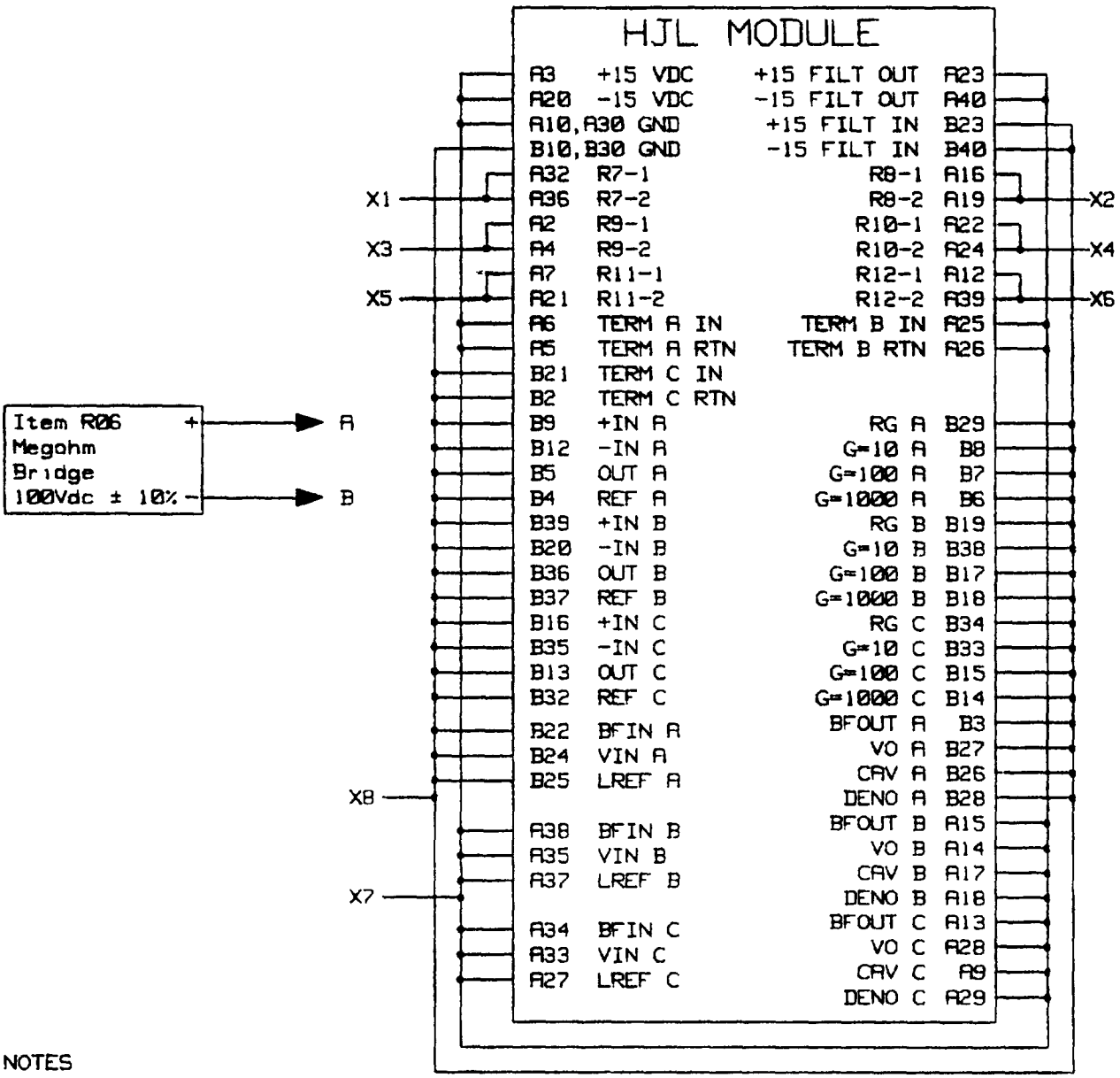
Test cond	Parameter	Connect		Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	Min	Max		
A1	Isolation, all pins to frame	A1,A8,A11,A31,B1, B11,B31,X1-X8	Frame	10	INF	MOhm	4
A2	Isolation, all pins	A1,A8,A11,A31,B1, B11,B31,X1	X2-X8				
A3	Isolation, all pins	A1,A8,A11,A31,X2, X3-X5	B1,B11,B31,X1, X6-X8				
A4	Isolation, all pins	A1,A8,X2,X3,B1, B11,X6,X7	A11,A31,X4,X5, B31,X1,X8				
A5	Isolation, all pins	A1,X2,B1,X6,A11, X4,B31,X8	A8,X3,B11,X7,X5, X1	10	INF	MOhm	4

NOTES

1 Tests A2-A5 ensure that all pins, used and unused, are isolated from all others.

2 Columns 3 and 4 in the above table imply that points A and B are connected to all the listed pins at one time

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NOTES

1. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

2. For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

FIGURE 4. Isolation test circuit for table II.

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TABLE III. Maximum stress test.

Test cond	Parameter	Initial limits 25 ± 5°C	Fig
B	Maximum stress	Module shall pass all remaining tests	5

NOTE:

1. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds and a maximum of 60 seconds.

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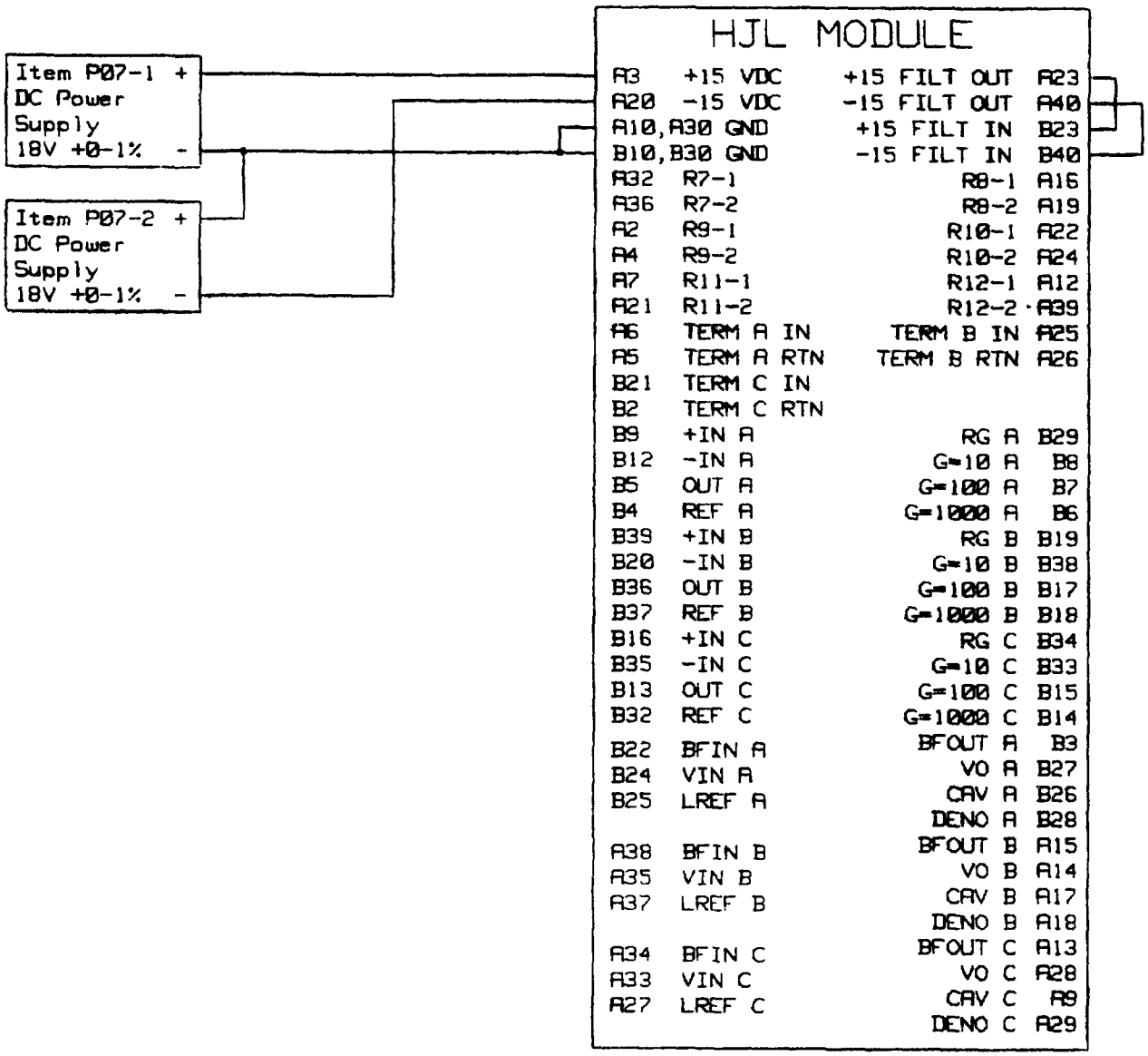


FIGURE 5 Maximum stress test circuit for table III

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TABLE IV. Output short circuit current test

Test cond	Parameter	Connect		Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	Min	Max		
C1	Short circuit current buffer A	B22	B3	10	40	mA	6
C2	Short circuit current buffer B	A38	A15				
C3	Short circuit current buffer C	A34	A13				
C4	Short circuit current RMS-DC A	B24	B27				
C5	Short circuit current RMS-DC B	A35	A14				
C6	Short circuit current RMS-DC C	A33	A28				
C7	Short circuit current RMS-DC A	B24	B27				
C8	Short circuit current RMS-DC B	A35	A14				
C9	Short circuit current RMS-DC C	A33	A28	10	40	mA	6

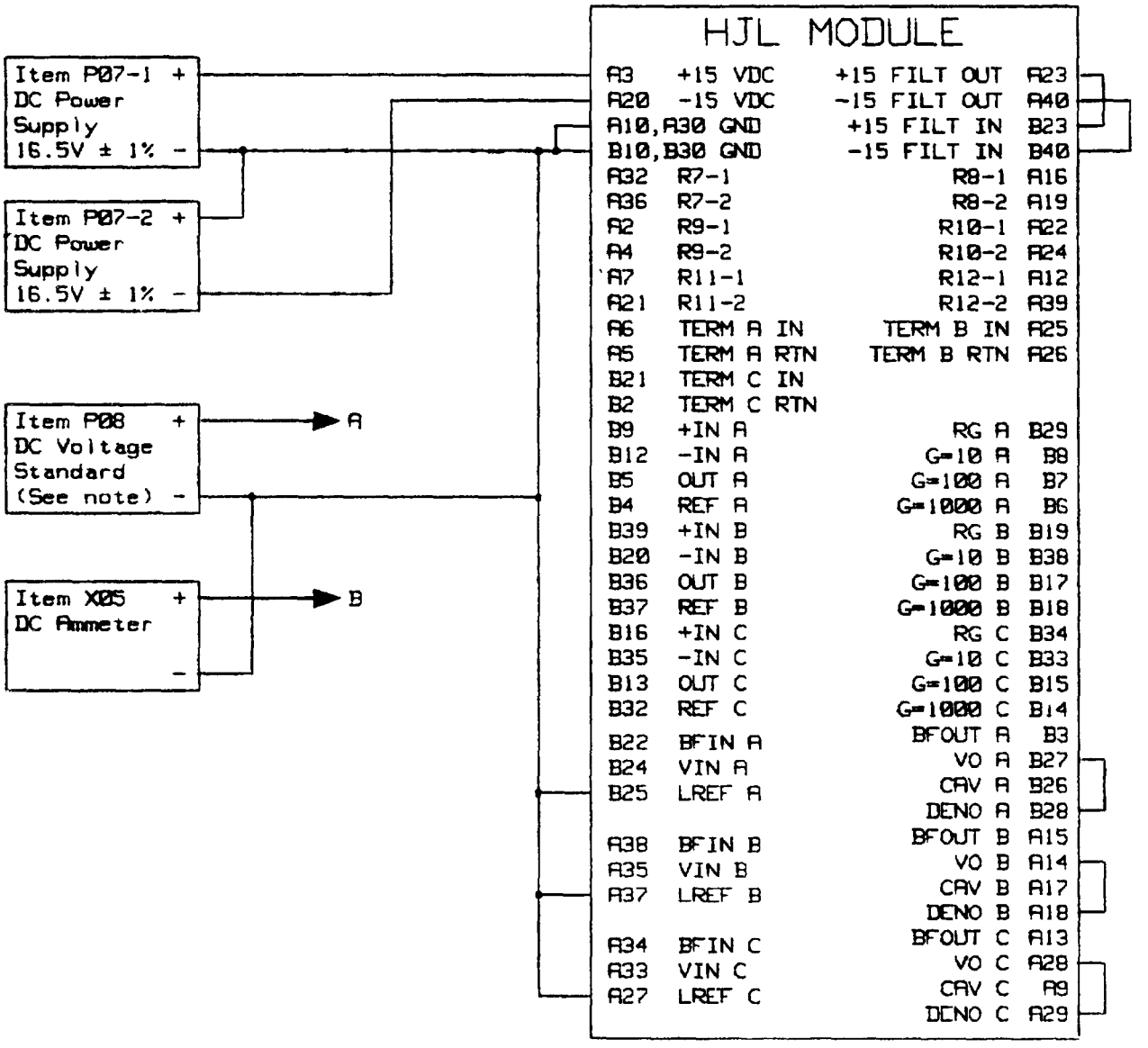
NOTES

1 With point A disconnected, item X05 should indicate zero current on the 1 uA range.

2 When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken after 1 minute.

3 The BAMP is not capable of delivering more than -130 uA.

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NOTE

1 For test conditions C1-C3 and C7-C9, Item P08 = 10 V ± 1 percent and for test conditions C4-C6, Item P08 = -10 V ± 1 percent

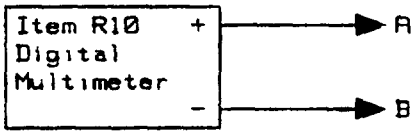
FIGURE 6 Output short circuit current test circuit for table IV

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TABLE V. Continuity test.

Test cond	Parameter	Connect		Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	Min	Max		
D1	Board A (signal/return/ground)	A10	A30	0.0	1.0	Ohm	7
D2	Board B (signal/return/ground)	B10	B30	0.0	1.0	Ohm	7
D3	Board A (signal/return/ground)	A10	A30	0.0	0.1	Ohm	7
D4	Board B (signal/return/ground)	B10	B30	0.0	0.1	Ohm	7

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HJL MODULE			
A3	+15 VDC	+15 FILT OUT	A23
A20	-15 VDC	-15 FILT OUT	A40
A10,A30	GND	+15 FILT IN	B23
B10,B30	GND	-15 FILT IN	B40
A32	R7-1	R8-1	A16
A36	R7-2	R8-2	A19
A2	R9-1	R10-1	A22
A4	R9-2	R10-2	A24
A7	R11-1	R12-1	A12
A21	R11-2	R12-2	A39
A6	TERM A IN	TERM B IN	A25
A5	TERM A RTN	TERM B RTN	A26
B21	TERM C IN		
B2	TERM C RTN		
B9	+IN A	RG A	B29
B12	-IN A	G=10 A	B8
B5	OUT A	G=100 A	B7
B4	REF A	G=1000 A	B6
B39	+IN B	RG B	B19
B20	-IN B	G=10 B	B38
B36	OUT B	G=100 B	B17
B37	REF B	G=1000 B	B18
B16	+IN C	RG C	B34
B35	-IN C	G=10 C	B33
B13	OUT C	G=100 C	B15
B32	REF C	G=1000 C	B14
B22	BFIN A	BFOUT A	B3
B24	VIN A	VO A	B27
B25	LREF A	CAV A	B26
		DENO A	B28
A38	BFIN B	BFOUT B	A15
A35	VIN B	VO B	A14
A37	LREF B	CAV B	A17
		DENO B	A18
A34	BFIN C	BFOUT C	A13
A33	VIN C	VO C	A28
A27	LREF C	CAV C	A9
		DENO C	A29

FIGURE 7 Continuity test circuit for table V

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TABLE VI. Power supply current test.

Test cond	Parameter	Initial limits 25 ± 5°C		Units	Fig
		Min	Max		
E1	Positive power supply current	1	24	mA	8
E2	Negative power supply current	1	24	mA	8

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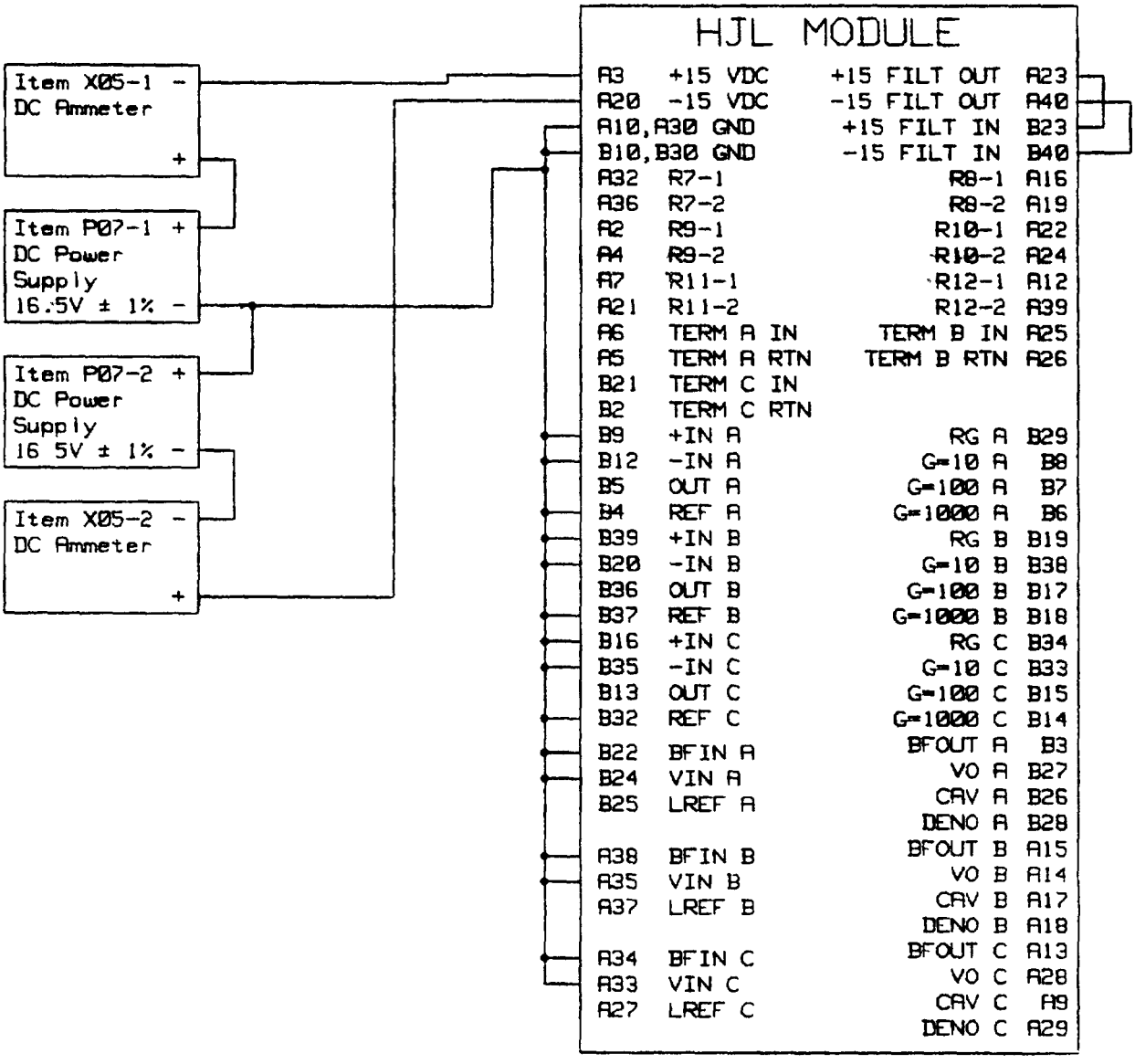


FIGURE 8 Power supply current test circuit for table VI

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TABLE VII. PSFC test.

Test cond	Parameter	Connect		Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	Min	Max		
F1	Board 1 positive Vcc filter capacitance	A23	A10,A30	2.07	2.53	uF	9
F2	Board 1 negative Vcc filter capacitance	A10,A30	A40	2.07	2.53	uF	9
F3	Board 2 positive Vcc filter capacitance	B10,B30	B23	0.36	0.44	uF	9
F4	Board 2 negative Vcc filter capacitance	B10,B30	B40	0.36	0.44	uF	9

NOTES

1 Capacitance bridge. Frequency = 1000 hertz, amplitude = 0.3 volts peak maximum.

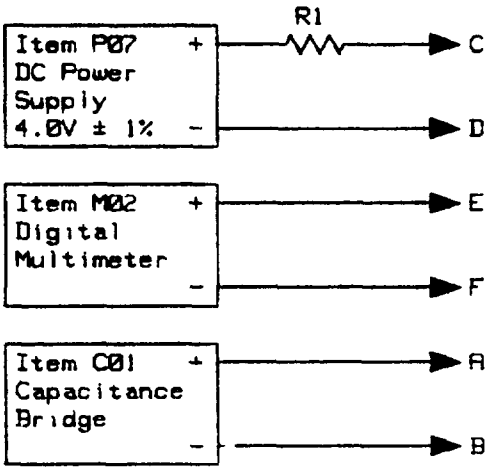
2 Null out the effects of the test leads before connecting to module.

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TABLE VIII. Diode voltage drop test.

Test cond	Parameter	Connect				Initial limits 25 ± 5°C		Units	Fig
		C to pin	D to pin	E to pin	F to pin	Min	Max		
G	Forward voltage drop	A20	A3	A20	A3	0.4	0.9	V	9

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HJL MODULE			
A3	+15 VDC	+15 FILT OUT	A23
A20	-15 VDC	-15 FILT OUT	A40
A10, A30	GND	+15 FILT IN	B23
B10, B30	GND	-15 FILT IN	B40
A32	R7-1	R8-1	A16
A36	R7-2	R8-2	A19
A2	R9-1	R10-1	A22
A4	R9-2	R10-2	A24
A7	R11-1	R12-1	A12
A21	R11-2	R12-2	A39
A6	TERM A IN	TERM B IN	A25
A5	TERM A RTN	TERM B RTN	A26
B21	TERM C IN		
B2	TERM C RTN		
B9	+IN A	RG A	B29
B12	-IN A	G=10 A	B8
B5	OUT A	G=100 A	B7
B4	REF A	G=1000 A	B6
B39	+IN B	RG B	B19
B20	-IN B	G=10 B	B38
B36	OUT B	G=100 B	B17
B37	REF B	G=1000 B	B18
B16	+IN C	RG C	B34
B35	-IN C	G=10 C	B33
B13	OUT C	G=100 C	B15
B32	REF C	G=1000 C	B14
B22	BFIN A	BFOUT A	B3
B24	VIN A	VO A	B27
B25	LREF A	CAV A	B26
		DENO A	B28
A38	BFIN B	BFOUT B	A15
A35	VIN B	VO B	A14
A37	LREF B	CAV B	A17
		DENO B	A18
A34	BFIN C	BFOUT C	A13
A33	VIN C	VO C	A28
A27	LREF C	CAV C	A9
		DENO C	A29

NOTES:

1. Pins A23, A40, B23 and B40 must be shorted to prevent component damage
2. R1 = 120 Ohm ± 0.1 percent, 0.5 W.

FIGURE 9 PSFC and diode voltage drop test circuit for tables VII and VIII

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TABLE IX. Inductor check test.

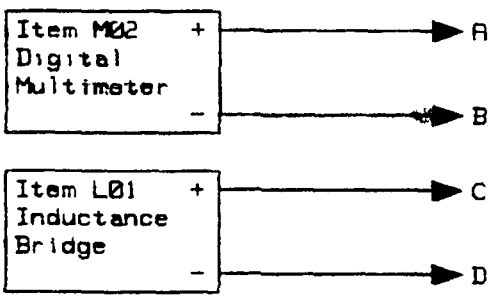
Test cond	Parameter	Connect				Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Resistance check	A3	A23			30.0	80.0	Ohm	10
H2	Resistance check	A40	A20			30.0	80.0	Ohm	10
H3	Inductance check			A3	A23	3.5	4.3	mH	10
H4	Inductance check			A40	A20	3.5	4.3	mH	10

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TABLE X. Resistance check test.

Test cond	Parameter	Connect		Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	Min	Max		
J1	R1	A5	A10,A30	37.03	37.77	Ohm	10
J2	R2	A6	A10,A30				
J3	R3	A26	A10,A30				
J4	R4	A25	A10,A30				
J5	R5	B21	B10,B30				
J6	R6	B2	B10,B30	37.03	37.77	Ohm	10
J7	R7	A32	A36	74.93	75.08	kOhm	10
J8	R8	A16	A19				
J9	R9	A2	A4				
J10	R10	A22	A24	74.93	75.08	kOhm	10
J11	R11	A7	A21	529.47	530.53	kOhm	10
J12	R12	A12	A39	529.47	530.53	kOhm	10

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HJL MODULE			
A3	+15 VDC	+15 FILT OUT	A23
A20	-15 VDC	-15 FILT OUT	A40
A10,A30	GND	+15 FILT IN	B23
B10,B30	GND	-15 FILT IN	B40
A32	R7-1	R8-1	A16
A36	R7-2	R8-2	A19
A2	R9-1	R10-1	A22
A4	R9-2	R10-2	A24
A7	R11-1	R12-1	A12
A21	R11-2	R12-2	A39
A6	TERM A IN	TERM B IN	A25
A5	TERM A RTN	TERM B RTN	A26
B21	TERM C IN		
B2	TERM C RTN		
B9	+IN A	RG A	B29
B12	-IN A	G=10 A	B8
B5	OUT A	G=100 A	B7
B4	REF A	G=1000 A	B6
B39	+IN B	RG B	B19
B20	-IN B	G=10 B	B38
B36	OUT B	G=100 B	B17
B37	REF B	G=1000 B	B18
B16	+IN C	RG C	B34
B35	-IN C	G=10 C	B33
B13	OUT C	G=100 C	B15
B32	REF C	G=1000 C	B14
B22	BFIN A	BFOUT A	B3
B24	VIN A	VO A	B27
B25	LREF A	CAV A	B26
		DENO A	B28
A38	BFIN B	BFOUT B	A15
A35	VIN B	VO B	A14
A37	LREF B	CAV B	A17
		DENO B	A18
A34	BFIN C	BFOUT C	A13
A33	VIN C	VO C	A28
A27	LREF C	CAV C	A9
		DENO C	A29

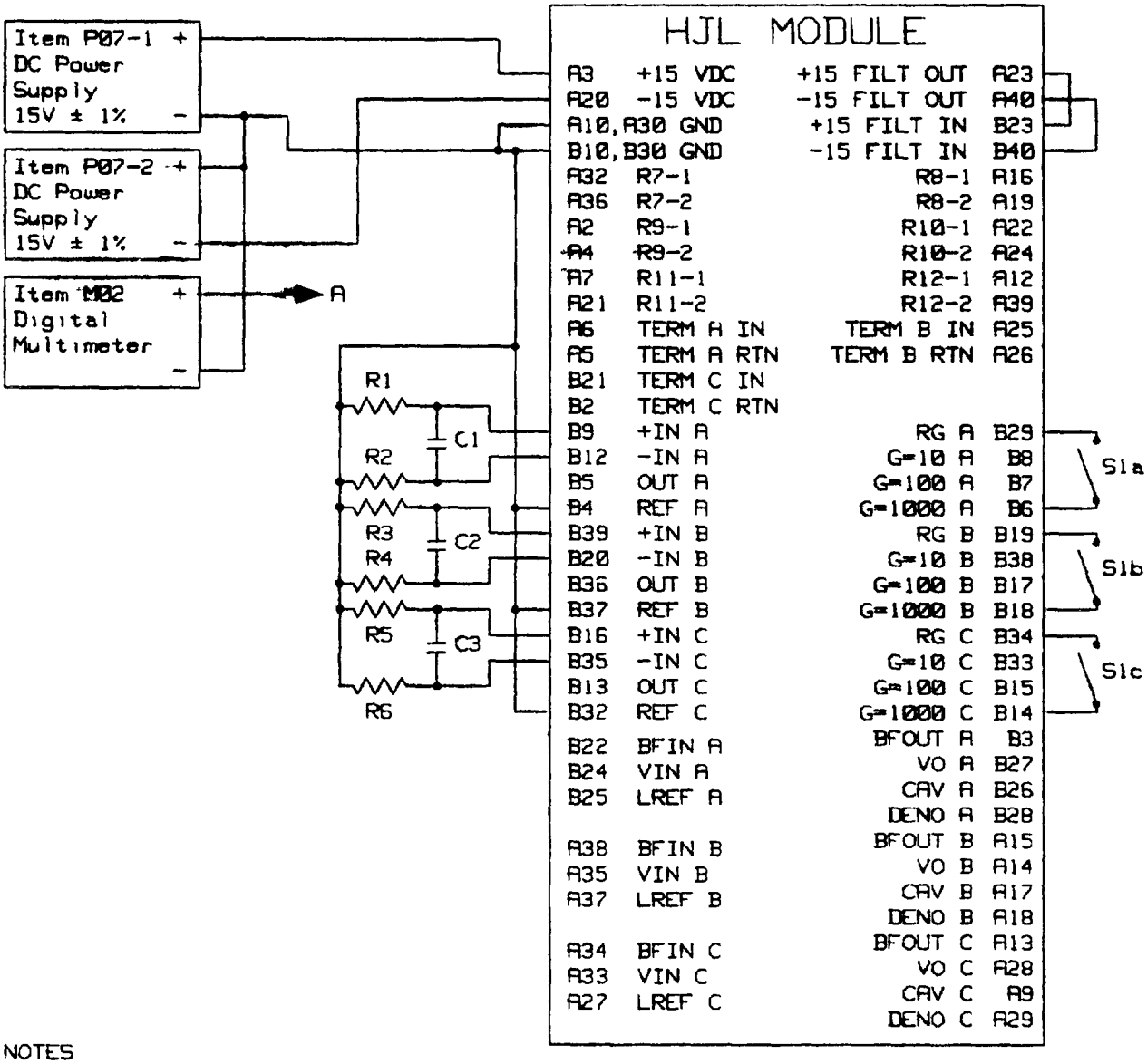
FIGURE 10 Inductor and resistance check test circuit for tables IX and X.

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TABLE XI. IAMP offset voltage test.

Test cond	Parameter	SI position	Connect A to pin	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
					Min	Max		
		Open	B5	#1				11
		Closed	B5	#2				11
K1	Input offset voltage IAMP A			K1=(#2-#1)/999	-110.0	110.0	uV	11
K2	Output offset voltage IAMP A			K2=(1000*#1-#2)/999	-3.25	3.25	mV	11
		Open	B36	#3				11
		Closed	B36	#4				11
K3	Input offset voltage IAMP B			K3=(#4-#3)/999	-110.0	110.0	uV	11
K4	Output offset voltage IAMP B			K4=(1000*#3-#4)/999	-3.25	3.25	mV	11
		Open	B13	#5				11
		Closed	B13	#6				11
K5	Input offset voltage IAMP C			K5=(#6-#5)/999	-110.0	110.0	uV	11
K6	Output offset voltage IAMP C			K6=(1000*#5-#6)/999	-3.25	3.25	mV	11

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NOTES

1. R1 = R2 = R3 = R4 = R5 = R6 = 100 Ohm ± 1 percent, 0.1 W
2. C1 = C2 = C3 = 0.1 uF ± 5 percent, 20 V
3. S1 = 3PST switch

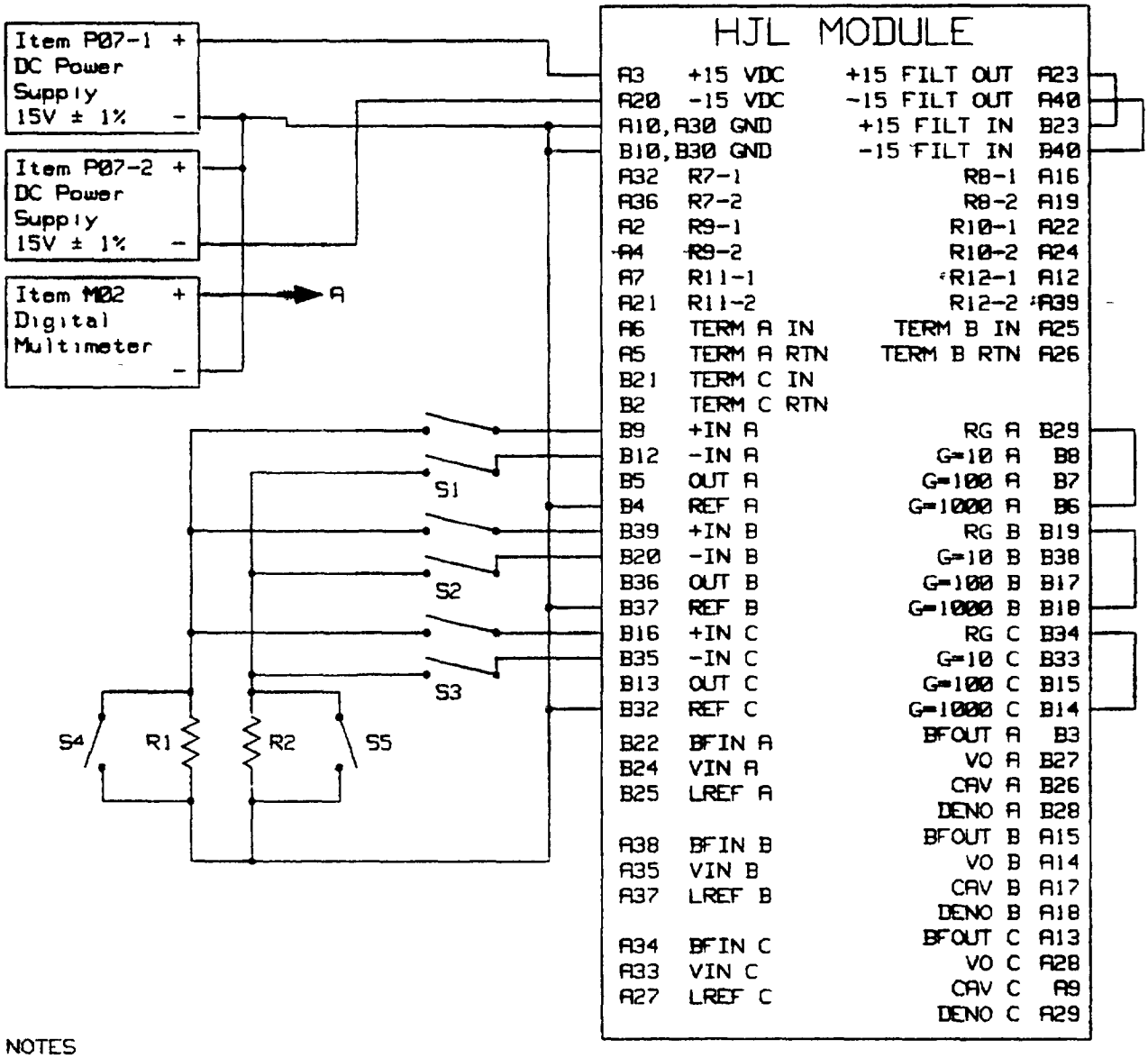
FIGURE 11. IAMP offset voltage test circuit for table XI

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TABLE XII. IAMP input bias and offset current test

Test cond	Parameter	Switches closed	Connect A to pin	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
					Min	Max		
		S1,S4,S5	B5	#1				12
		S1,S5	|	#2				|
		S1,S4	B5	#3				12
L1	Input bias current Ib(+) IAMP A			L1=(#2-#1)/100,000,000	-50.5	50.5	nA	12
L2	Input bias current Ib(-) IAMP A			L2=(#1-#3)/100,000,000	-50.5	50.5	nA	12
L3	Input offset current Ios IAMP A			L3=L2-L1	-35.5	35.5	nA	12
		S2,S4,S5	B36	#4				12
		S2,S5	|	#5				|
		S2,S4	B36	#6				12
L4	Input bias current Ib(+) IAMP B			L4=(#5-#4)/100,000,000	-50.5	50.5	nA	12
L5	Input bias current Ib(-) IAMP B			L5=(#4-#6)/100,000,000	-50.5	50.5	nA	12
L6	Input offset current Ios IAMP B			L6=L5-L4	-35.5	35.5	nA	12
		S3,S4,S5	B13	#7				12
		S3,S5	|	#8				|
		S3,S4	B13	#9				12
L7	Input bias current Ib(+) IAMP C			L7=(#8-#7)/100,000,000	-50.5	50.5	nA	12
L8	Input bias current Ib(-) IAMP C			L8=(#7-#9)/100,000,000	-50.5	50.5	nA	12
L9	Input offset current Ios IAMP C			L9=L8-L7	-35.5	35.5	nA	12

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NOTES

1. R1 = R2 = 100 kOhm ± 1 percent, 0.1 W
2. S1 = S2 = S3 = DPST switch.
3. S4 = S5 = SPST switch

FIGURE 12. IAMP input bias and offset current test circuit for table XII.

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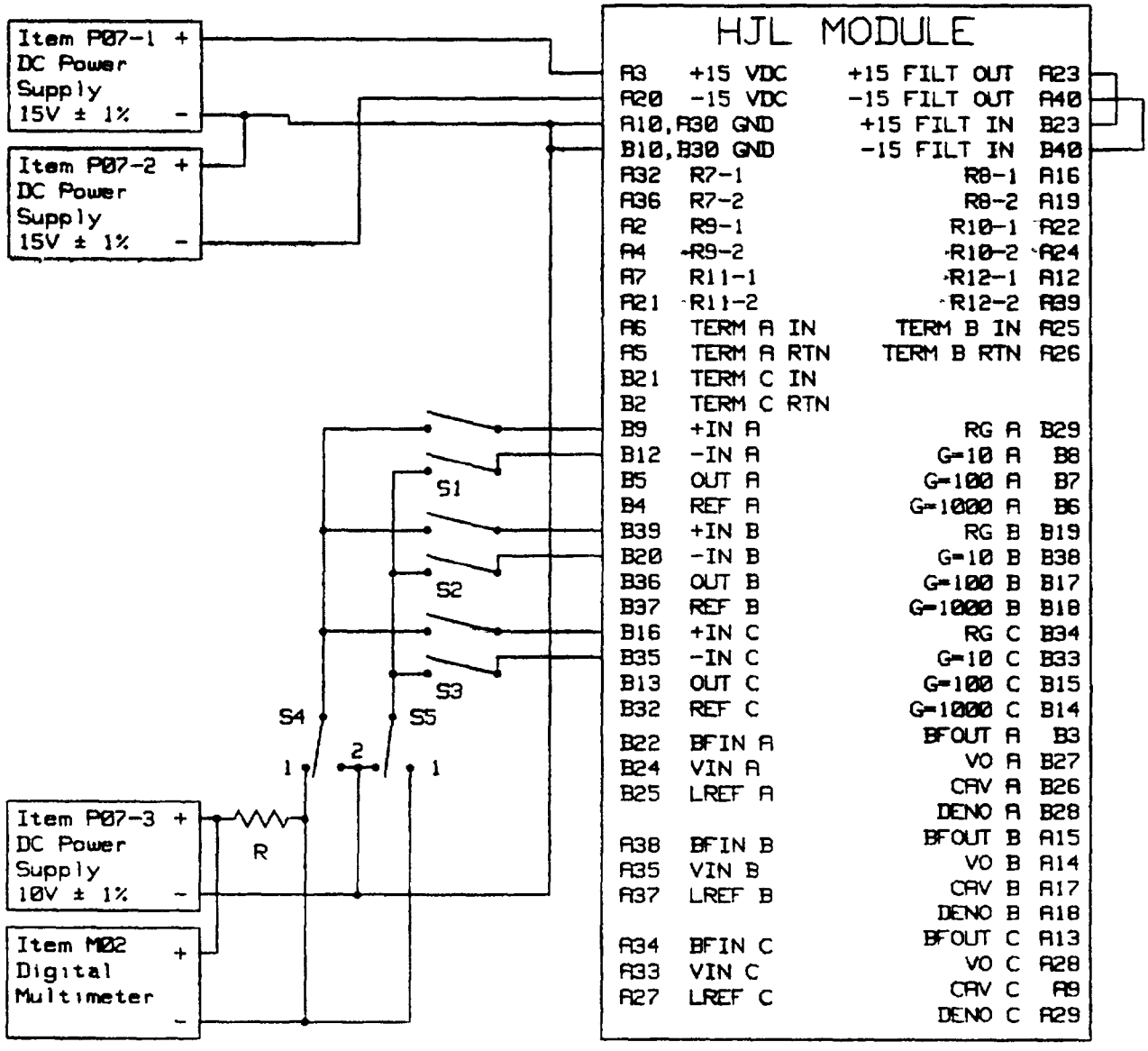
TABLE XIII. IAMP input resistance test.

Test cond	Parameter	Switches closed	Switch position		Initial limits 25 ± 5°C		Units	Fig
			S4	S5	Min	Max		
M1	Difference mode input resistance (+) IAMP A	S1	1	2	0	918	mV	13
M2	Difference mode input resistance (-) IAMP A	S1	2	1				
M3	Difference mode input resistance (+) IAMP B	S2	1	2				
M4	Difference mode input resistance (-) IAMP B	S2	2	1				
M5	Difference mode input resistance (+) IAMP C	S3	1	2				
M6	Difference mode input resistance (-) IAMP C	S3	2	1				
M7	Common mode input resistance IAMP A	S1	1	1				
M8	Common mode input resistance IAMP B	S2	1	1				
M9	Common mode input resistance IAMP C	S3	1	1	0	918	mV	13

NOTE

1 The limit of 918 mV guarantees a minimum differetial and common mode input resistance of 10 MOhms.

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NOTES:

1. R = 1 MOhm ± 1 percent, 0.25 W.
2. S1 = S2 = S3 = DPST switch.
3 S4 = S5 = SPDT switch.

FIGURE 13 IAMP input resistance test circuit for table XIII

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TABLE XIV. IAMP gain accuracy test.

Test cond	Parameter	Switch closed	Atten. setting	Connect		Item P08	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
				A to pin	B to pin			Min	Max		
			g=1	B12	B5	0 V	#1				14
			g=1	B12	B5	10 V	#2				14
N1	Gain accuracy						N1=[5-(#2-#1)]/(g*5)	0.9985	1 0015	V/V	14
			g=1	B20	B36	0 V	#3				14
			g=1	B20	B36	10 V	#4				14
N2	Gain accuracy						N2=[5-(#4-#3)]/(g*5)	0 9985	1.0015	V/V	14
			g=1	B35	B13	0 V	#5				14
			g=1	B35	B13	10 V	#6				14
N3	Gain accuracy						N3=[5-(#6-#5)]/(g*5)	0.9985	1.0015	V/V	14
		S1	g=0 1	B12	B5	0 V	#7				14
		S1	g=0 1	B12	B5	10 V	#8				14
N4	Gain accuracy						N4=[5-(#8-#7)]/(g*5)	9.975	10 025	V/V	14
		S1	g=0 1	B20	B36	0 V	#9				14
		S1	g=0 1	B20	B36	10 V	#10				14
N5	Gain accuracy						N5=[5-(#10-#9)]/(g*5)	9 975	10 025	V/V	14
		S1	g=0 1	B35	B13	0 V	#11				14
		S1	g=0 1	B35	B13	10 V	#12				14
N6	Gain accuracy						N6=[5-(#12-#11)] /(g*5)	9 975	10 025	V/V	14
		S2	g=0.01	B12	B5	0 V	#13				14
		S2	g=0.01	B12	B5	10 V	#14				14
N7	Gain accuracy						N7=[5-(#14-#13)]/(g*5)	99.5	100 5	V/V	14
		S2	g=0.01	B20	B36	0 V	#15				14
		S2	g=0.01	B20	B36	10 V	#16				14

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TABLE XIV IAMP gain accuracy test - Continued.

Test cond	Parameter	Switch closed	Atten setting	Connect		Item P08	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
				A to pin	B to pin			Min	Max		
N8	Gain accuracy						N8=[5-(#16-#15)]/(g*5)	99.5	100 5	V/V	14
		S2	g=0 01	B35	B13	0 V	#17				14
		S2	g=0 01	B35	B13	10 V	#18				14
N9	Gain accuracy						N9=[5-(#18-#17)]/(g*5)	99.5	100.5	V/V	14
		S3	g=0 001	B12	B5	0 V	#19				14
		S3	g=0.001	B12	B5	10 V	#20				14
N10	Gain accuracy						N10=[5-(#20-#19)]/(g*5)	980 0	1020.0	V/V	14
		S3	g=0 001	B20	B36	0 V	#21				14
		S3	g=0 001	B20	B36	10 V	#22				14
N11	Gain accuracy						N11=[5-(#22-#21)]/(g*5)	980 0	1020.0	V/V	14
		S3	g=0 001	B35	B13	0 V	#23				14
		S3	g=0.001	B35	B13	10 V	#24				14
N12	Gain accuracy						N12=[5-(#24-#23)]/(g*5)	980.0	1020.0	V/V	14

NOTES

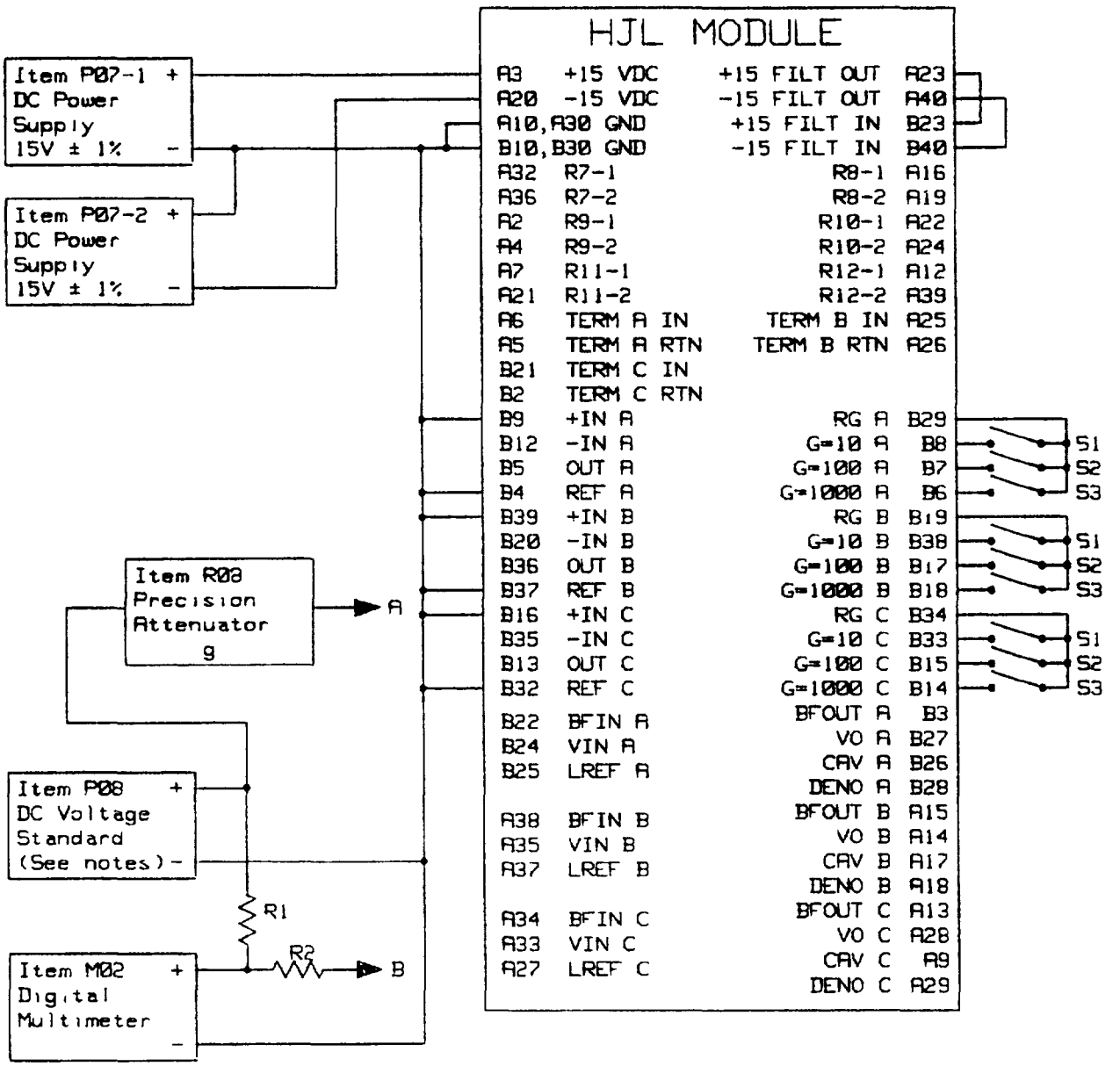
1. Testing shall be performed as follows

(a) After satisfying the setup conditions, make the specified measurement corresponding to the test condition.

(b) For those test conditions not requiring a measurement, perform the specified computation.

2. The gains of the attenuator (g) must be within ± 10 percent of the reciprocal of the appropriate amplifier gain. Those gains must be known to within ± 0.1 percent.

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NOTES

1 R1 = R2 = 49 9 kOhm, matched to 0 005 percent, 0 25 W

2 S1 = S2 = S3 = 3PST switch

3 Adjust Item P08 to values in table XIV ± 0 1 V

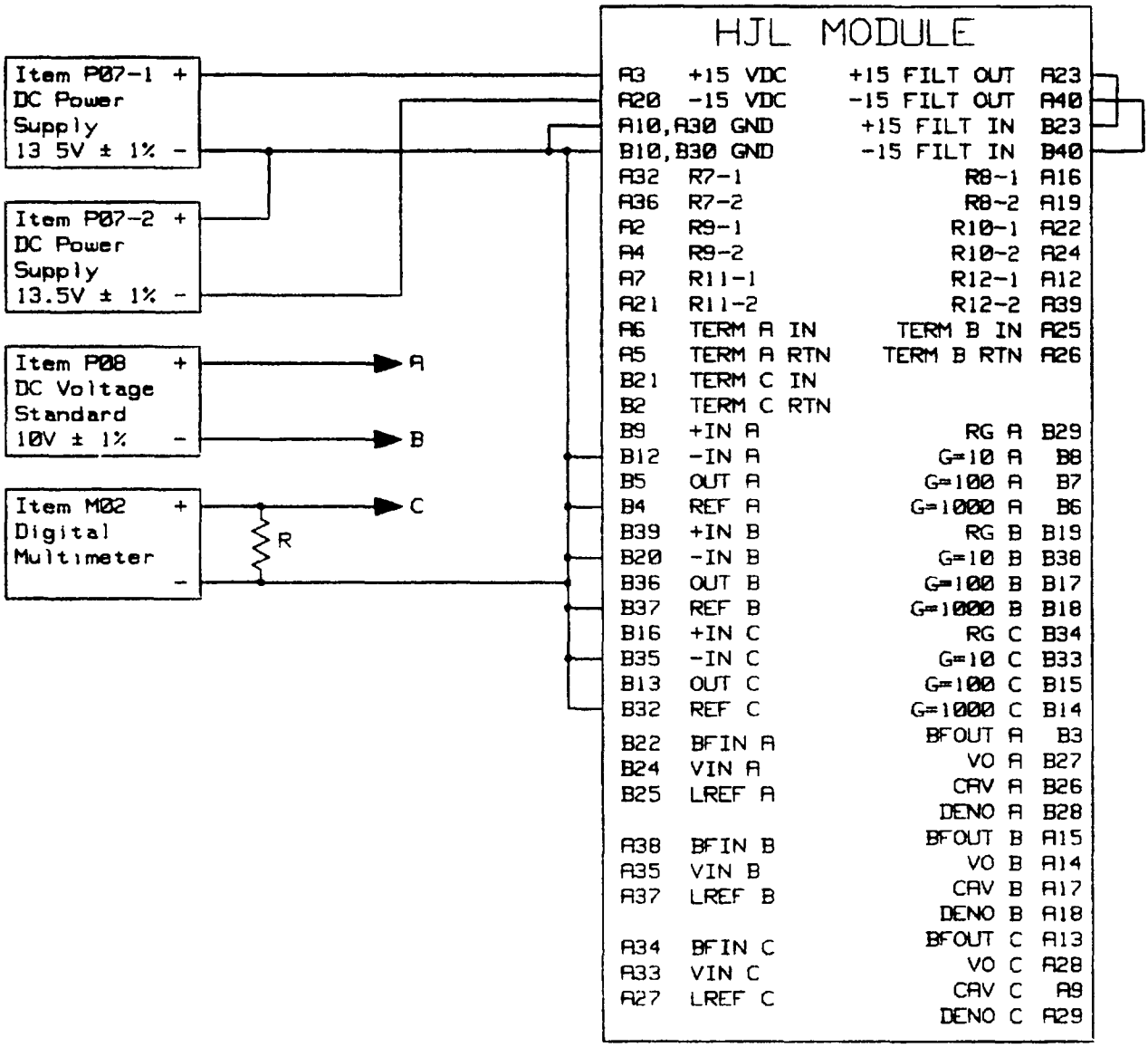
FIGURE 14 IAMP gain accuracy test circuit for table XIV

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TABLE XV. Output voltage swing test

Test cond	Parameter	Connect			Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin	C to pin	Min	Max		
P1	Output voltage swing	B9	A10	B5	9.5	11.0	V	15
P2	"	B39	A10	B36	"	"	"	"
P3	"	B16	A10	B13	9.5	11.0	V	15
P4	"	A10	B9	B5	-11.0	-9.5	V	15
P5	"	A10	B39	B36	"	"	"	"
P6	Output voltage swing	A10	B16	B13	-11.0	-9.5	V	15

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NOTE:

1 R = 2 0 kOhm ± 1 percent, 0 25 W

FIGURE 15 Output voltage swing test circuit for table XV

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TABLE XVI IAMP CMRR test.

Test cond	Parameter	Connect A to pin	Switch closed	Item P08 (Vdc) ± 1 percent	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
						Min	Max		
		B5		10	#1				16
		B5		-10	#2				16
Q1	CMRR G=1 IAMP A				$Q1=20*LOG\frac{20}{\#1-\#2}$	70	INF	dB	16
		B36		10	#3				16
		B36		-10	#4				16
Q2	CMRR G=1 IAMP B				$Q2=20*LOG\frac{20}{\#3-\#4}$	70	INF	dB	16
		B13		10	#5				16
		B13		-10	#6				16
Q3	CMRR G=1 IAMP C				$Q3=20*LOG\frac{20}{\#5-\#6}$	70	INF	dB	16
		B5	S1	10	#7				16
		B5	S1	-10	#8				16
Q4	CMRR G=10 IAMP A				$Q4=20*LOG\frac{200}{\#7-\#8}$	90	INF	dB	16
		B36	S1	10	#9				16
		B36	S1	-10	#10				16
Q5	CMRR G=10 IAMP B				$Q5=20*LOG\frac{200}{\#9-\#10}$	90	INF	dB	16
		B13	S1	10	#11				16
		B13	S1	-10	#12				16
Q6	CMRR G=10 IAMP C				$Q6=20*LOG\frac{200}{\#11-\#12}$	90	INF	dB	16
		B5	S2	10	#13				16
		B5	S2	-10	#14				16

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TABLE XVI. IAMP CMRR test - Continued.

Test cond	Parameter	Connect A to pin	Switch closed	Item P08 (Vdc) ± 1 percent	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
						Min	Max		
Q7	CMRR G=100 IAMP A				$Q7=20*LOG\frac{2000}{\#13-\#14}$	100	INF	dB	16
		B36	S2	10	#15				16
		B36	S2	-10	#16				16
Q8	CMRR G=100 IAMP B				$Q8=20*LOG\frac{2000}{\#15-\#16}$	100	INF	dB	16
		B13	S2	10	#17				16
		B13	S2	-10	#18				16
Q9	CMRR G=100 IAMP C				$Q9=20*LOG\frac{2000}{\#17-\#18}$	100	INF	dB	16
		B5	S3	10	#19				16
		B5	S3	-10	#20				16
Q10	CMRR G=1000 IAMP A				$Q10=20*LOG\frac{20000}{\#19-\#20}$	110	INF	dB	16
		B36	S3	10	#21				16
		B36	S3	-10	#22				16
Q11	CMRR G=1000 IAMP B				$Q11=20*LOG\frac{20000}{\#21-\#22}$	110	INF	dB	16
		B13	S3	10	#23				16
		B13	S3	-10	#24				16
Q12	CMRR G=1000 IAMP C				$Q12=20*LOG\frac{20000}{\#23-\#24}$	110	INF	dB	16

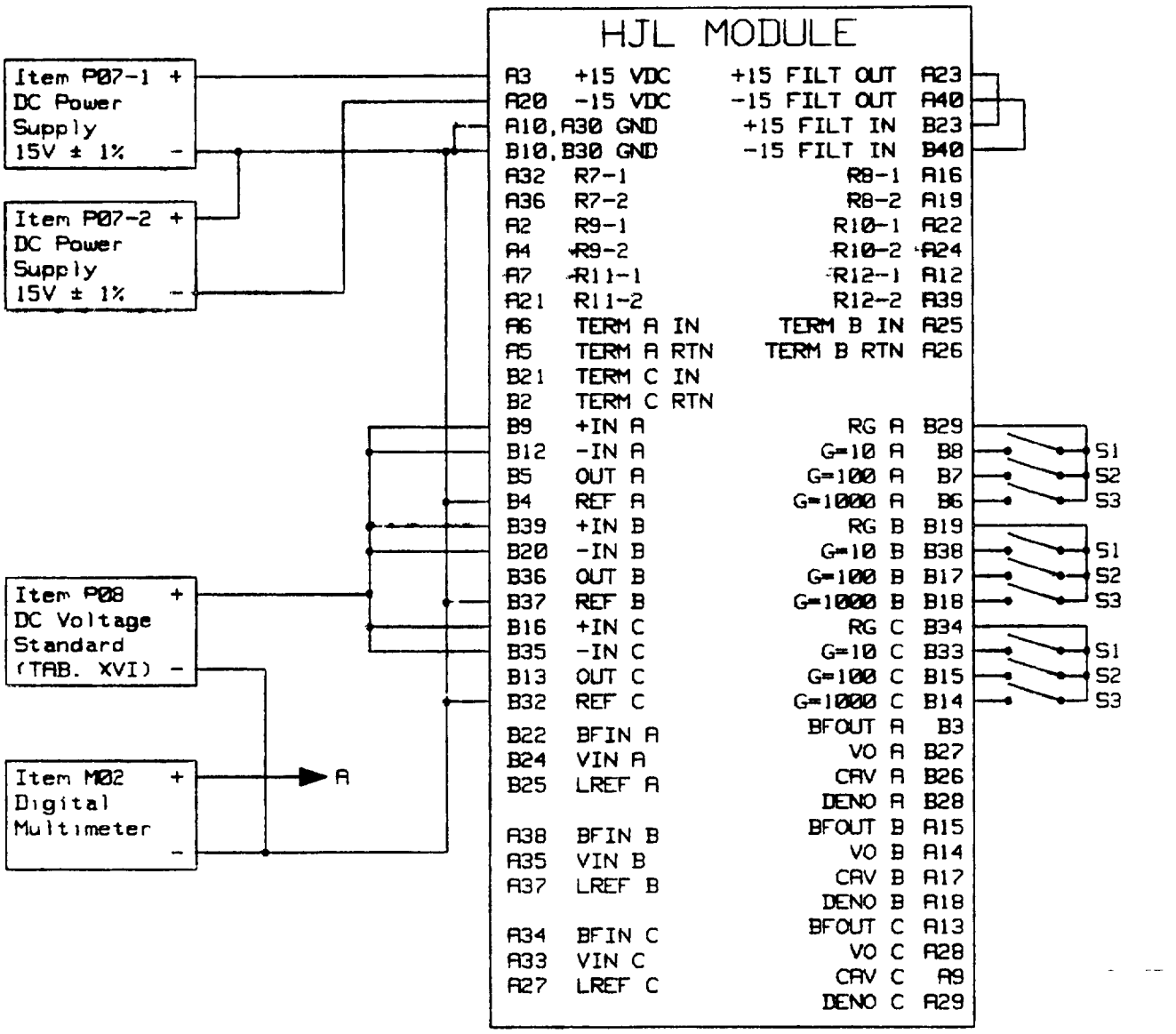
NOTE

1 Testing shall be performed as follows

(a) After satisfying the setup conditions, make the specified measurement corresponding to the test conditions.

(b) For those test conditions not requiring a measurement, perform the specified computation.

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NOTE:

1. S1 = S2 = S3 = 3PST switch

FIGURE 16 IAMP CMRR test circuit for table XVI.

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TABLE XVII. IAMP PSS test

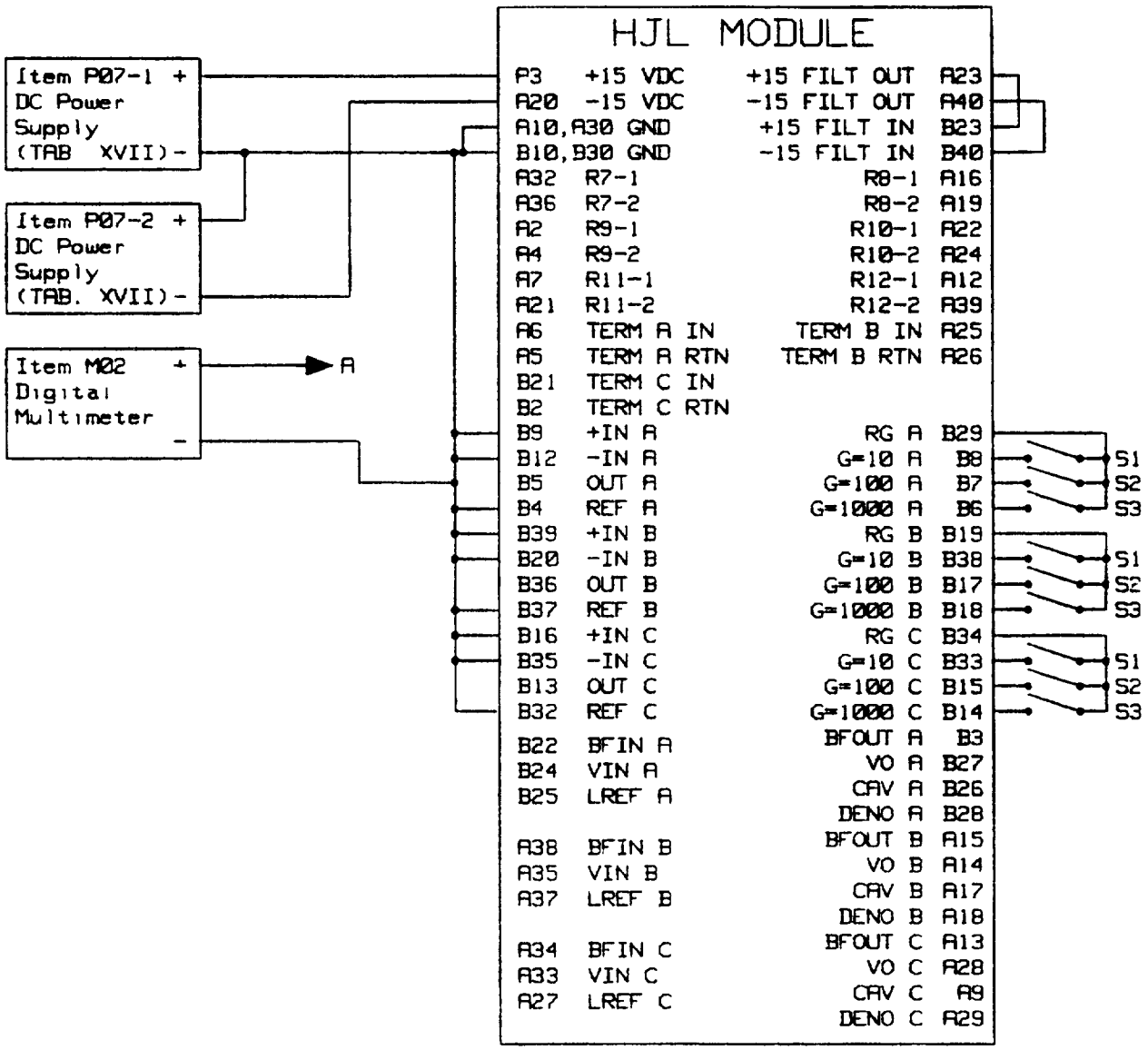
Test cond	Parameter	Item P07-1 (Vac) ± 1 percent	Item P07-2 (Vdc) ± 1 percent	Connect A to pin	Switch closed	Measurement/ computation	Initial limits 25 ± 5°C Min	Initial limits 25 ± 5°C Max	Units	Fig
		16 5	-16.5	B5		#1				17
		|	|	B36		#2				|
		16 5	-16 5	B13		#3				|
		13 5	-13.5	B5		#4				|
		|	|	B36		#5				|
		13 5	-13 5	B13		#6				17
R1	PSS G=1 IAMP A					R1=20*LOG 6/(#1-#4)	75	INF	dB	17
R2	PSS G=1 IAMP B					R2=20*LOG 6/(#2-#5)	|	|	|	|
R3	PSS G=1 IAMP C					R3=20*LOG 6/(#3-#6)	75	INF	dB	17
		16.5	-16.5	B5	S1	#7				17
		|	|	B36	|	#8				|
		16 5	-16.5	B13	|	#9				|
		13 5	-13.5	B5	|	#10				|
		|	|	B36	|	#11				|
		13 5	-13.5	B13	S1	#12				17
R4	PSS G=10 IAMP A					R4=20*LOG 60/(#7-#10)	95	INF	dB	17
R5	PSS G=10 IAMP B					R5=20*LOG 60/(#8-#11)	|	|	|	|
R6	PSS G=10 IAMP C					R6=20*LOG 60/(#9-#12)	95	INF	dB	17

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TABLE XVII. IAMP PSS test - Continued.

Test cond	Parameter	Item P07-1 (Vdc) ± 1 percent	Item P07-2 (Vdc) ± 1 percent	Connect A to pin	Switch closed	Measurement/ computation	Initial limits 25 ± 5°C Min	Initial limits 25 ± 5°C Max	Units	Fig
		16.5	-16.5	B5	S2	#13				17
		|	|	B36	|	#14				|
		16.5	-16.5	B13	|	#15				|
		13.5	-13.5	B5	|	#16				|
		|	|	B36	|	#17				|
		13.5	-13.5	B13	S2	#18				17
R7	PSS G=100 IAMP A					R7=20*LOG 600/(#13-#16)	105	INF	dB	17
R8	PSS G=100 IAMP B					R8=20*LOG 600/(#14-#17)	|	|	|	|
R9	PSS G=100 IAMP C					R9=20*LOG 600/(#15-#18)	105	INF	dB	17
		16.5	-16.5	B5	S3	#19				17
		|	|	B36	|	#20				|
		16.5	-16.5	B13	|	#21				|
		13.5	-13.5	B5	|	#22				|
		|	|	B36	|	#23				|
		13.5	-13 5	B13	S3	#24				17
R10	PSS G=1000 IAMP A					R10=20*LOG 6000/(#19-#22)	110	INF	dB	17
R11	PSS G=1000 IAMP B					R11=20*LOG 6000/(#20-#23)	|	|	|	|
R12	PSS G=1000 IAMP C					R12=20*LOG 6000/(#21-#24)	110	INF	dB	17

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NOTE:

1. S1 = S2 = S3 = 3PST switch

FIGURE 17. IAMP PSS test circuit for table XVII.

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TABLE XVIII. IAMP settling time test.

Test cond	Parameter	Switch closed	Item R08 atten.	Connect		Initial limits 25 ± 5°C		Units	Figs
				A to pin	B to pin	Min	Max		
S1	Settling time		1.0	B12	B5	0	90	uS	18,19
S2	|		|	B20	B36	|	|	|	|
S3	|		1.0	B35	B13	0	90	uS	18,19
S4	|	S1	0.1	B12	B5	0	30	uS	18,19
S5	|	|	|	B20	B36	|	|	|	|
S6	|	S1	0.1	B35	B13	|	|	|	|
S7	|	S2	0.01	B12	B5	|	|	|	|
S8	|	|	|	B20	B36	|	|	|	|
S9	|	S2	0.01	B35	B13	0	30	uS	18,19
S10	|	S3	0.001	B12	B5	0	150	uS	18,19
S11	|	|	|	B20	B36	|	|	|	|
S12	Settling time	S3	0.001	B35	B13	0	150	uS	18,19

NOTES:

1. Adjust R3 so that the tails of the output waveform in figure 19 align

2. Determine the time it takes for the output waveform to come to within 1.0 mV of its final value.

3. A 1X oscilloscope probe with a maximum capacitance of 10 pF shall be used

4 Settling time is measured for both positive and negative input transitions as shown in figure 19.

5. The oscilloscope shall be a waveform analyzer type capable of 100 averages per minute with a plug-in bandwidth minimum of 20 MHz (e.g., Tek 7854)

6 Item R08 requirements:

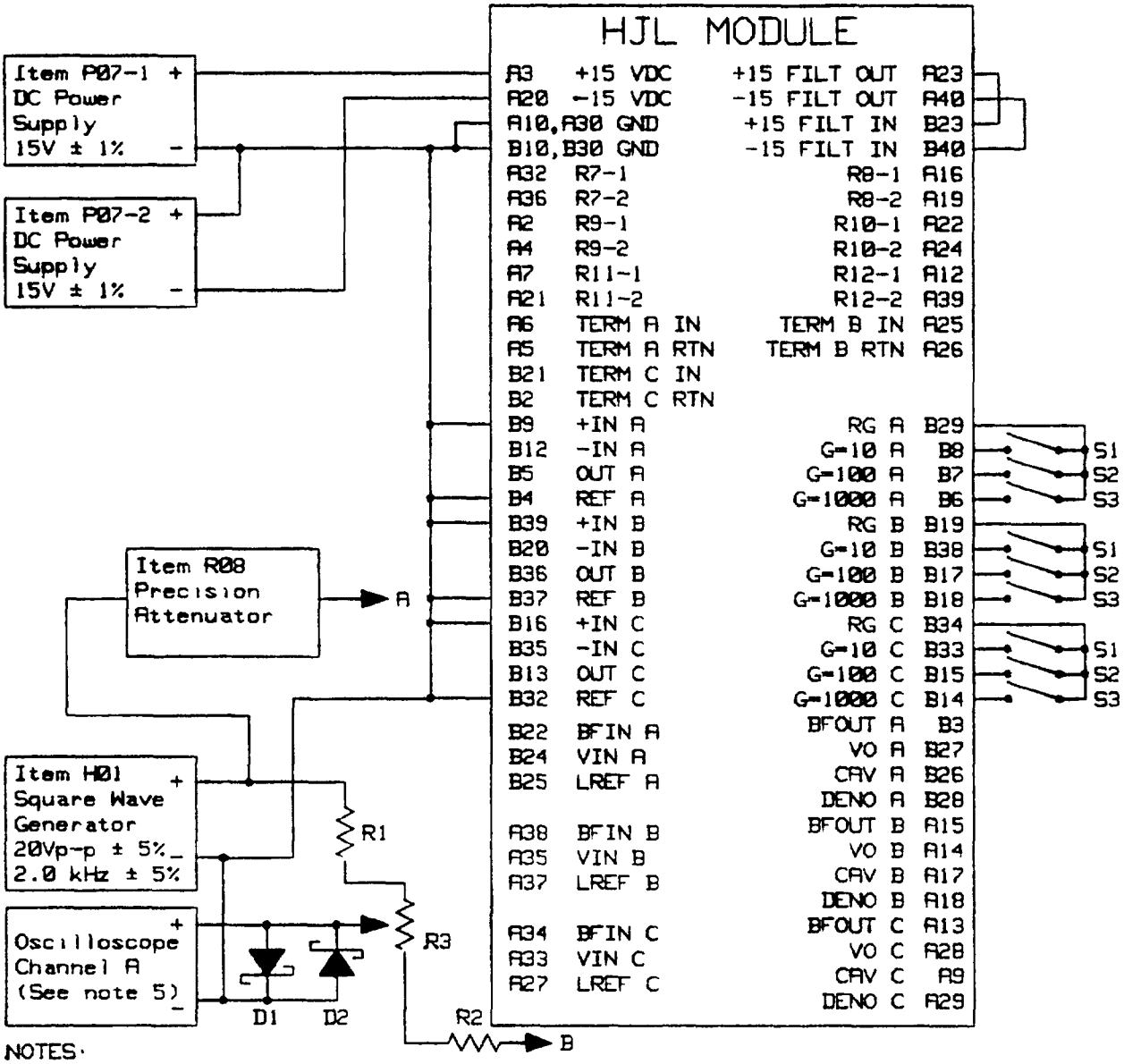
(a) g = 1.0 ± 0.05 percent

(b) g = 0.1 ± 0.25 percent

(c) g = 0.01 ± 0.50 percent

(d) g = 0.001 ± 2.00 percent

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NOTES:

1. S1 = S2 = S3 = 3PST switch
2. D1 = D2 = Schottky diodes.
3. R1 = R2 = 10.0 kOhm ± 1 percent, 0.25 W
4. R3 = 1.0 kOhm ± 5 percent, 20-turn pot, 0 25 W

FIGURE 18 IAMP settling time test circuit for table XVIII

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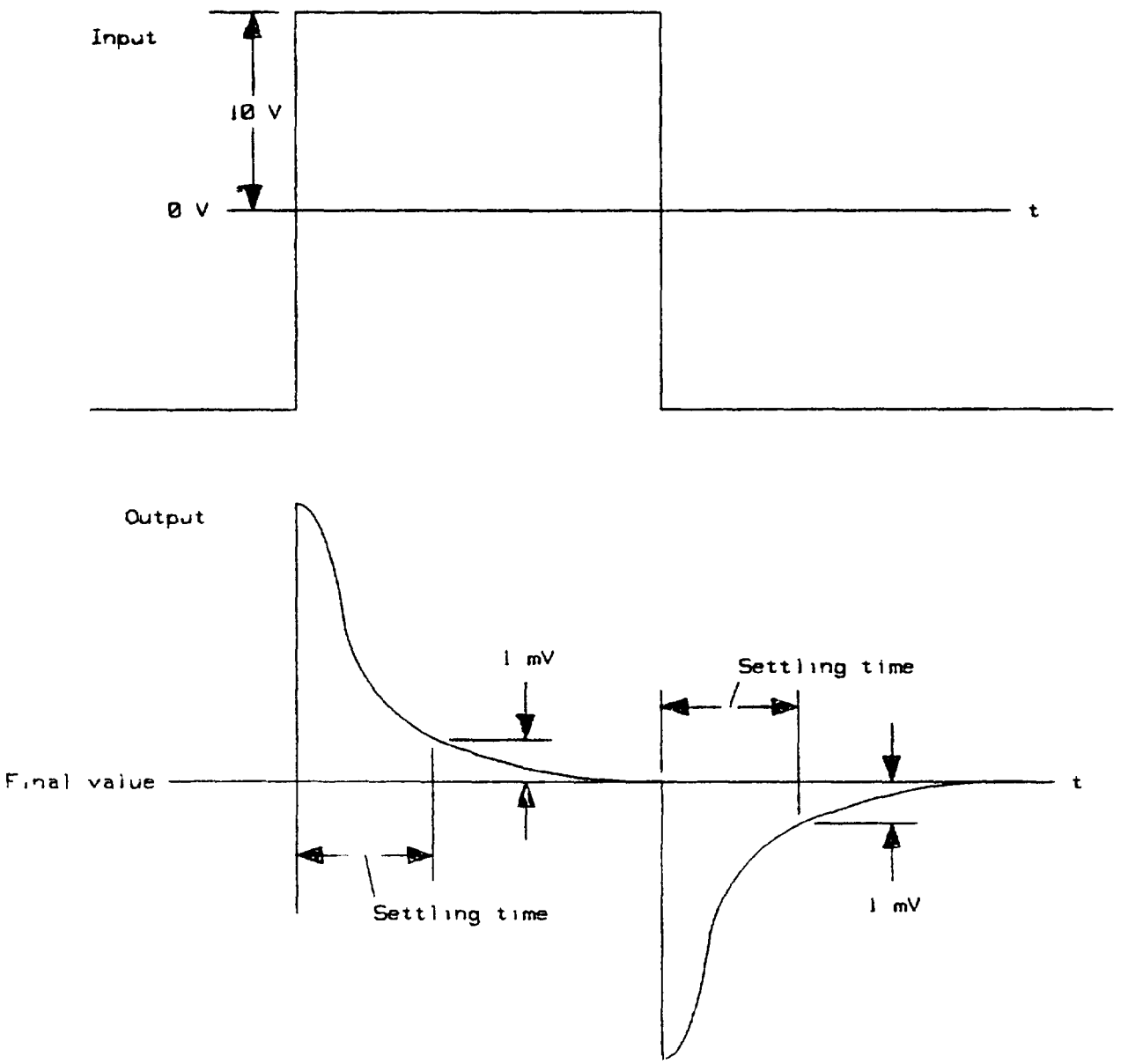


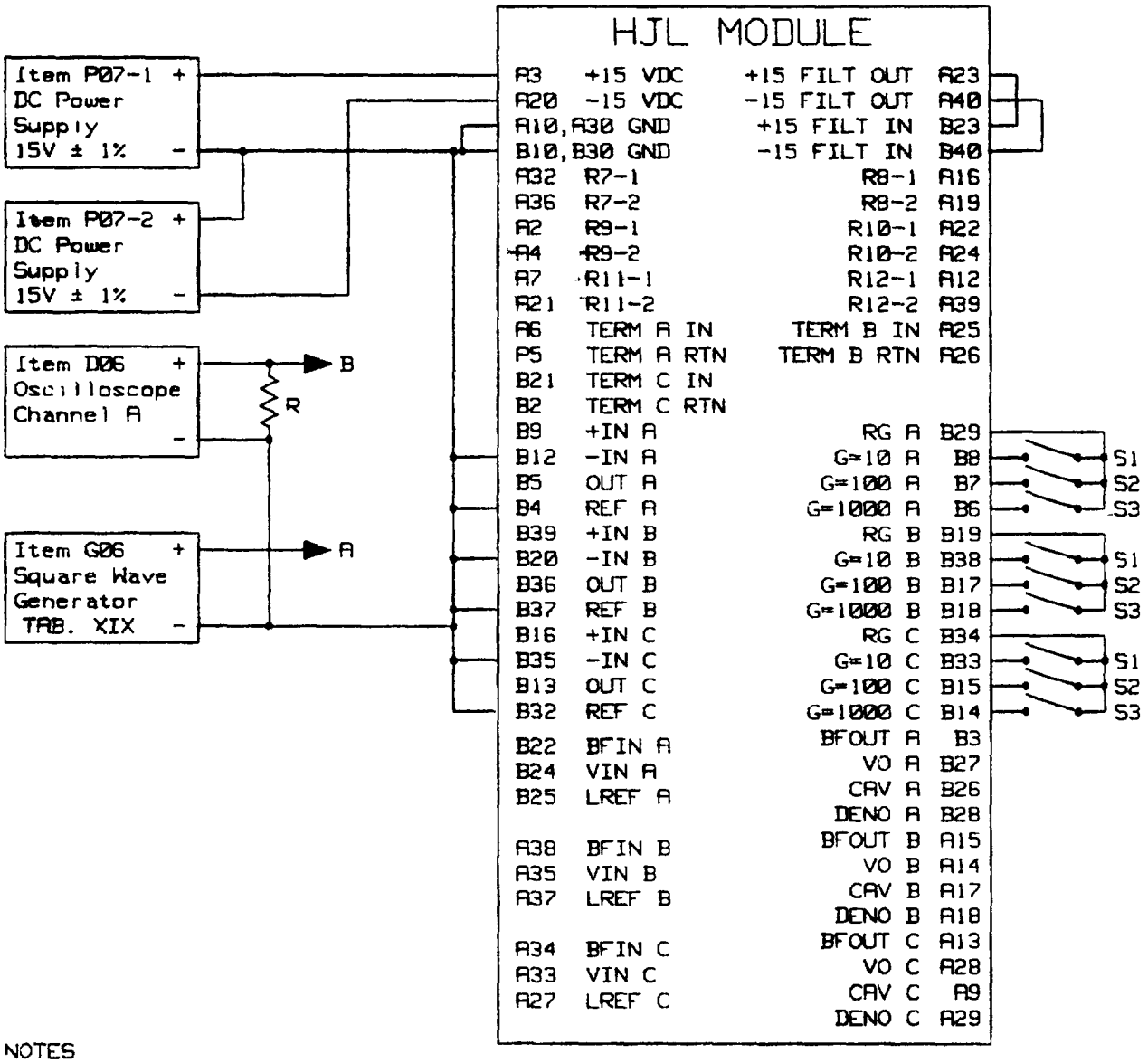
FIGURE 19 Waveforms for settling time test for figure 18

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TABLE XIX. IAMP slew rate test

Test cond	Parameter	Switch closed	Item G06 (Vpeak) ± 5 percent	Connect		Initial limits 25 ± 5°C		Units	Fig
				A to pin	B to pin	Min	Max		
T1	Slew rate		10.0	B9	B5	3.0	INF	V/uS	20,21
T2				B39	B36				
T3			10.0	B16	B13				
T4		S1	1.0	B9	B5				
T5				B39	B36				
T6		S1	1 0	B16	B13				
T7		S2	0.1	B9	B5				
T8				B39	B36				
T9		S2	0 1	B16	B13	3 0	INF	V/uS	20,21
T10		S3	0.01	B9	B5	0.5	INF	V/uS	20,21
T11				B39	B36				
T12	Slew rate	S3	0.01	B16	B13	0.5	INF	V/uS	20,21

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NOTES

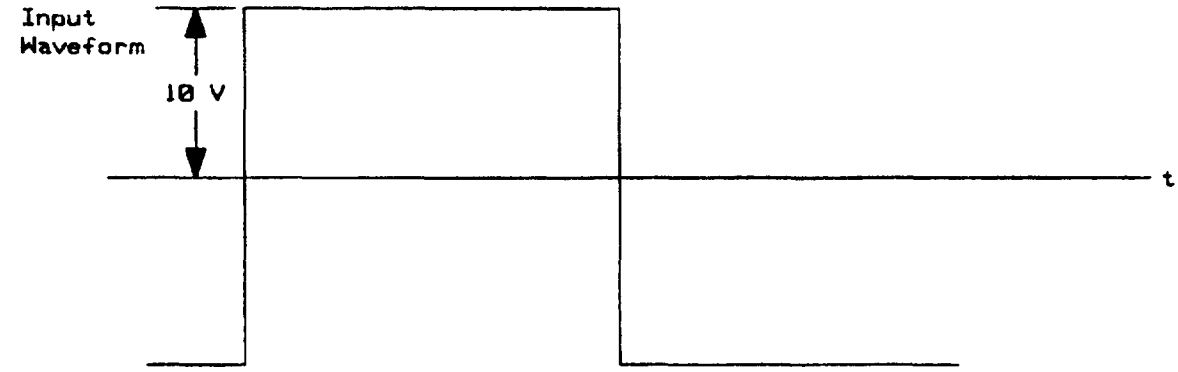
1 Scope probe = Y02

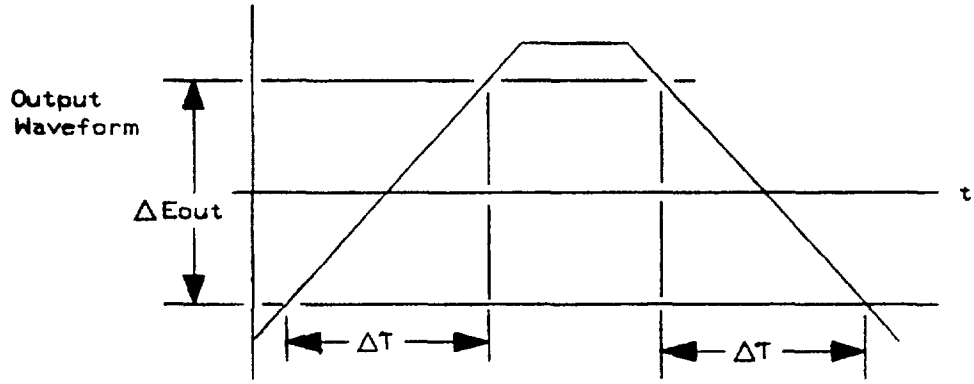
2 S1 = S2 = S3 = 3PST switch.

3 R = 2 kOhm ± 1 percent, 0 25 W

FIGURE 20. IAMP slew rate test circuit for table XIX.

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NOTES

1 For each test condition·

(a) Set the square wave generator to provide a 20 Vp-p square wave at a frequency of 60 kHz for T1-T9 and 3 kHz for T10-T12. The output waveform will be similar to that shown

(b) Determine both ΔT's as shown above Ensuring both ΔT's are below 3 3 uS will guarantee a slew rate of 3 V/uS.

(c) Compute slew rate as·

$$SR = \frac{\Delta Eout}{\Delta T} = \frac{10}{\Delta T}$$

FIGURE 21. Waveforms for slew rate test for figure 20

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TABLE XX. BAMP offset voltage test

Test cond	Parameter	Switch closed	Connect A to pin	Initial limits 25 ± 5°C		Units	Fig
				Min	Max		
U1	Input offset voltage	S1,S4	B3	-2	2	mV	22
U2	|	S2,S4	A15	|	|	|	|
U3	Input offset voltage	S3,S4	A13	-2	2	mV	22

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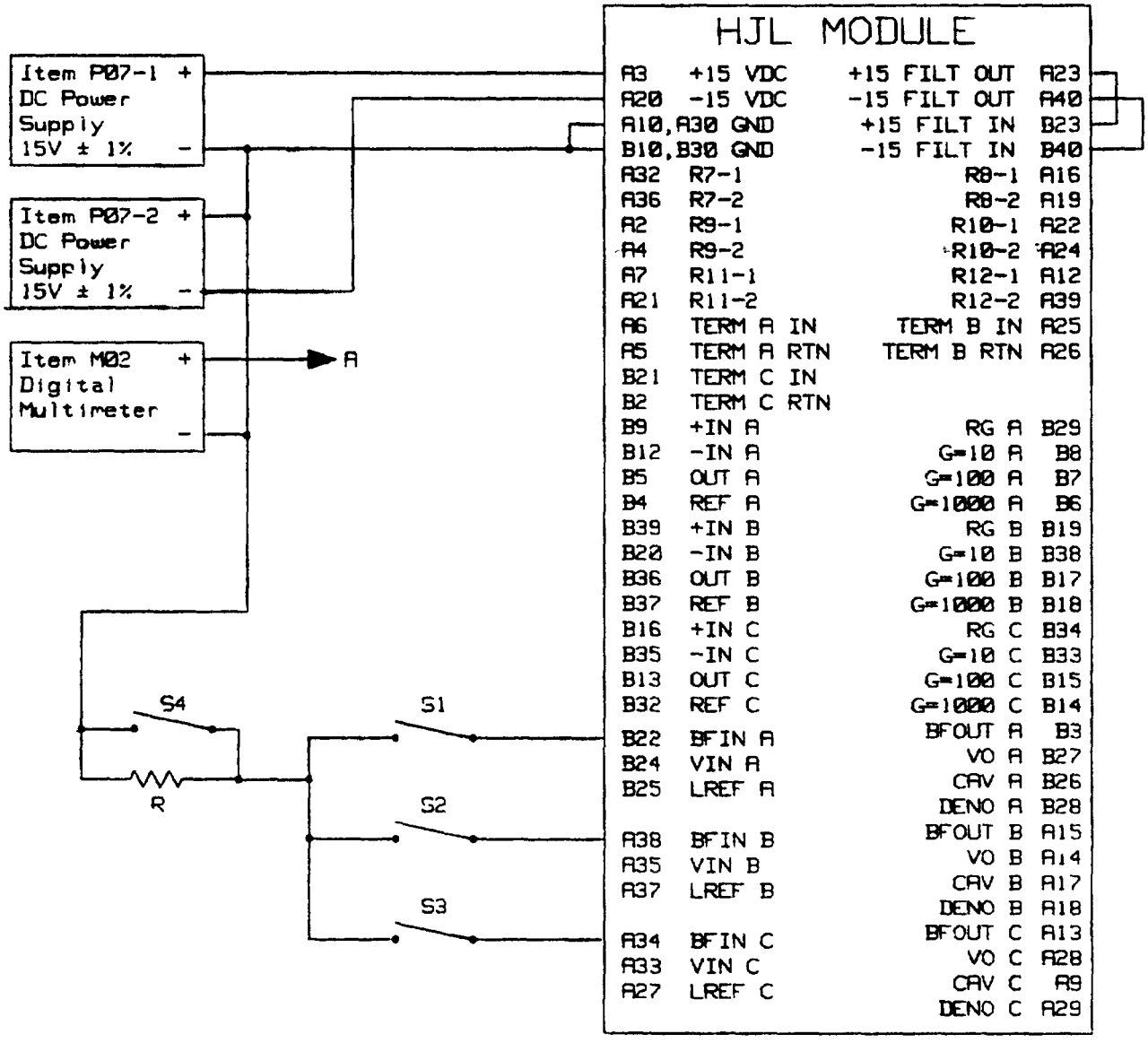
TABLE XXI. BAMP bias current test.

Test cond	Parameter	Switch closed	Connect A to pin	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
					Min	Max		
		S1	B3	#1				22
V1	Input bias current			V1=(U1-#1)/200000	-10	10	nA	22
		S2	A15	#2				22
V2	Input bias current			V2=(U2-#2)/200000	-10	10	nA	22
		S3	A13	#3				22
V3	Input bias current			V3=(U3-#3)/200000	-10	10	nA	22

NOTE

1 Test conditions U1-U3 are obtained from the buffer amplifier input offset in table XX.

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NOTES

1 S1 = S2 = S3 = S4 = SPST switch

2 R = 200 kOhm ± 1 percent, 0.25 W.

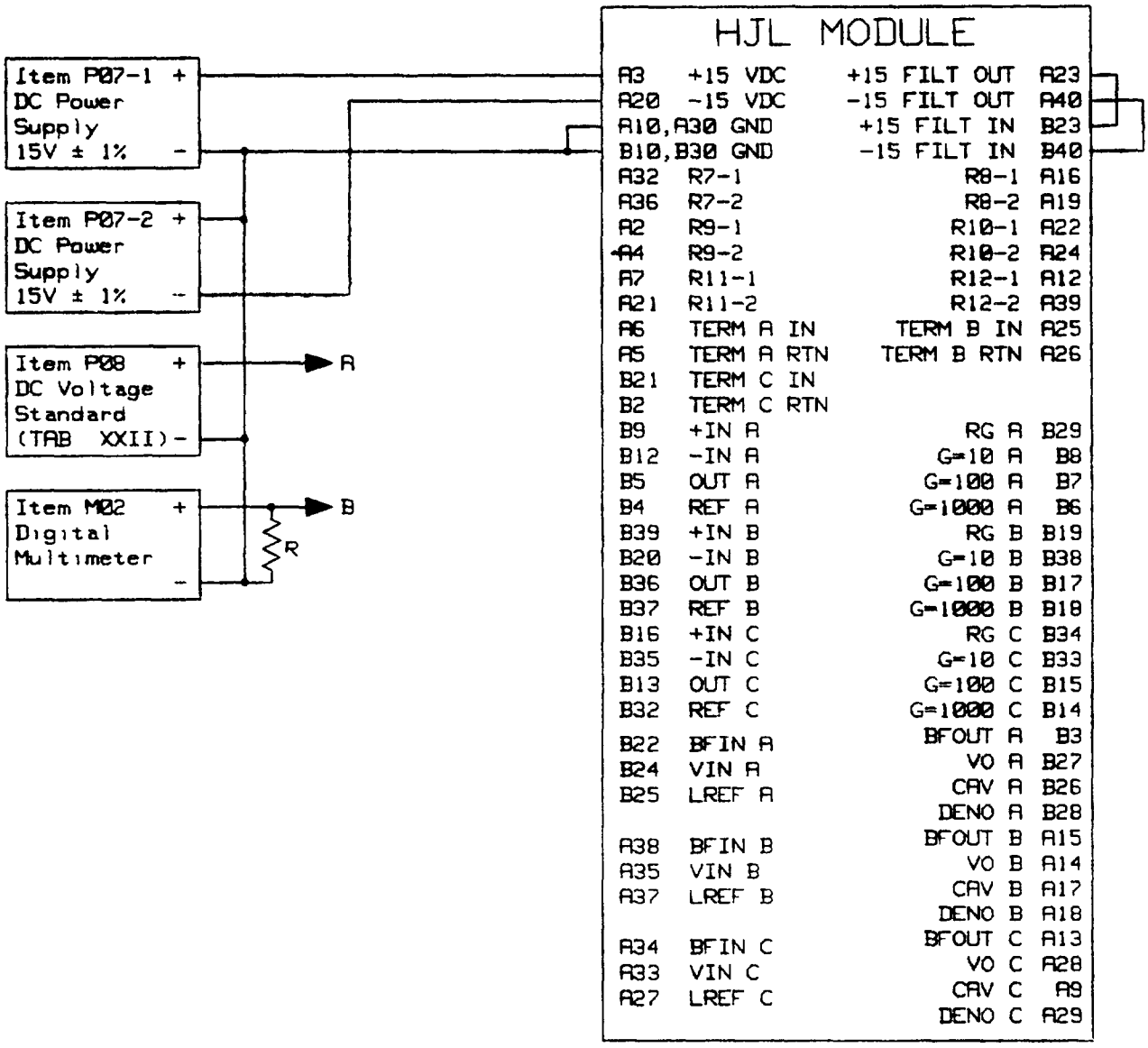
FIGURE 22. BAMP offset voltage and bias current test circuit for tables XX and XXI

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TABLE XXII. BAMP voltage swing test.

Test cond	Parameter	Connect		Item P08 (volts) ± 1 percent	Initial limits 25 ± 5°C		Units	Fig
		A to pin	B to pin		Min	Max		
W1	Voltage swing	B22	B3	10.0	9.0	11.0	V	23
W2	|	A38	A15	|	|	|	|	|
W3	Voltage swing	A34	A13	10 0	9.0	11.0	V	23

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NOTE

1 R = 2 kOhm ± 1 percent, 0.25 W

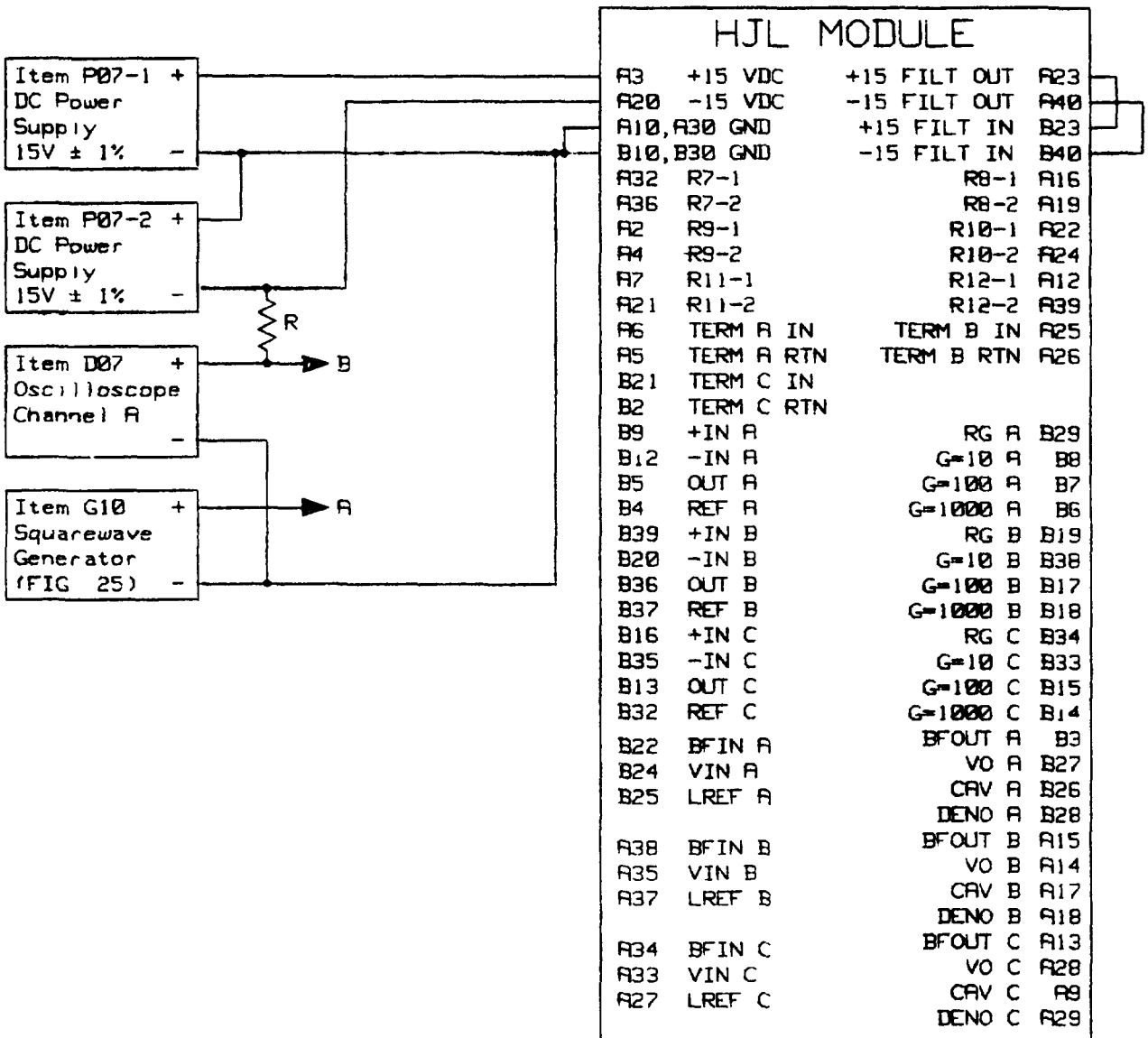
FIGURE 23 BAMP voltage swing test circuit for table XXII

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TABLE XXIII. BAMP slew rate test.

Test cond	Parameter	Connect		Initial limits 25 ± 5°C		Units	Figs
		A to pin	B to pin	Min	Max		
X1	Slew rate	B22	B3	3.75	INF	V/uS	24,25
X2	|	A38	A15	|	|	|	|
X3	Slew rate	A34	A13	3.75	INF	V/uS	24,25

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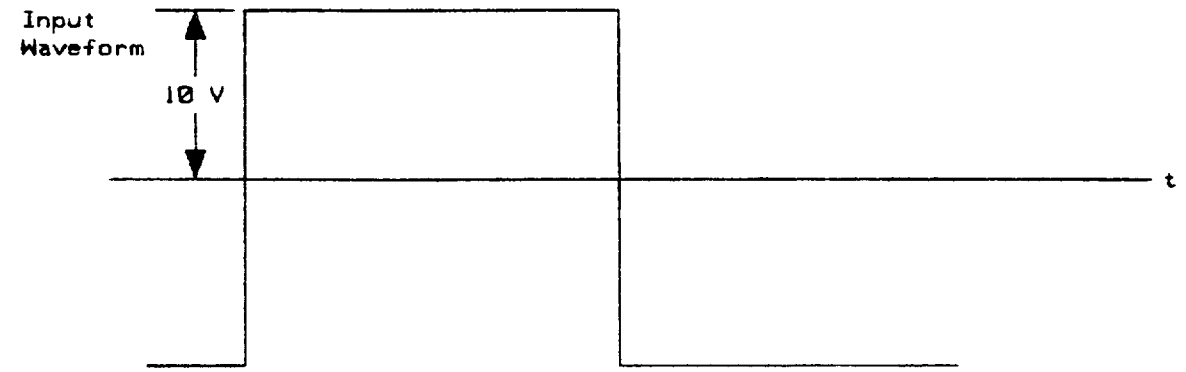
NOTES

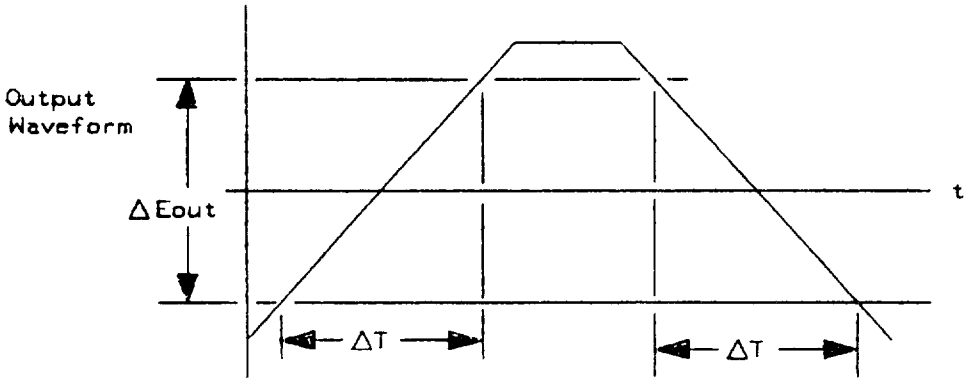
1 Scope probe = Y02

2 R = 2 kOhm ± 1 percent, 0.25 W

FIGURE 24 9AMP slew rate test circuit for table XXIII

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NOTES

1 For each test condition:

(a) Set the square wave generator to provide a 20 kHz, 20 Vp-p square wave The output waveform will be similar to that shown

(b) Determine both ΔT's as shown above Ensuring both ΔT's are below 2 7 uS will guarantee a minimum slew rate of 3 75 V/uS

(c) Compute slew rate as

$$SR = \frac{\Delta Eout}{\Delta T} = \frac{10}{\Delta T}$$

FIGURE 25 Waveforms for slew rate test for figure 24

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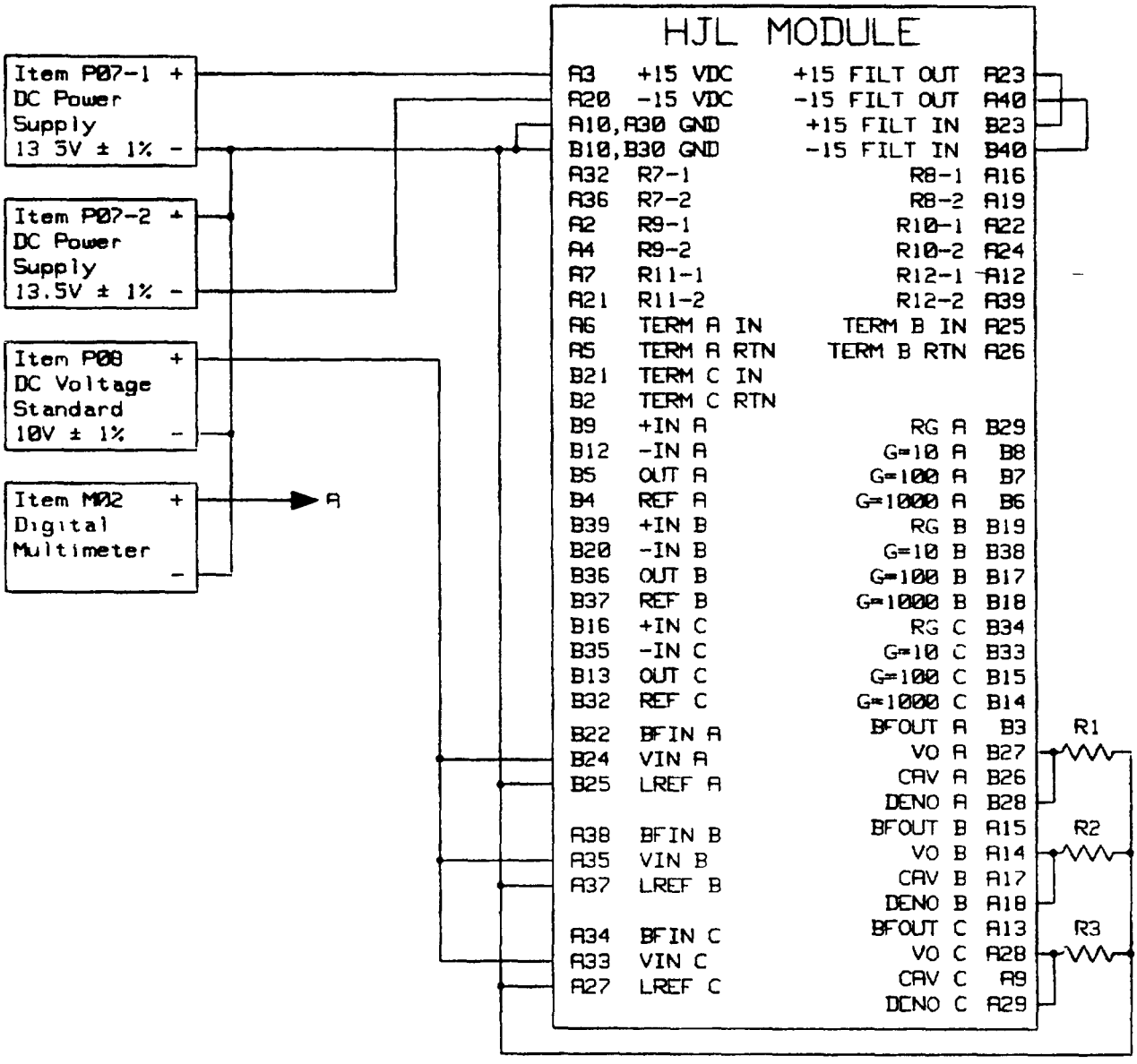
TABLE XXIV Output voltage swing test

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C		Units	Fig
			Min	Max		
Y1	Output voltage swing	B27	9.5	11 0	V	26
Y2	|	A14	|	|	|	|
Y3	Output voltage swing	A28	9.5	11.0	V	26

NOTE:

1 Item P08 voltage must not exceed 13.5 V dc

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NOTE

1 R1 = R2 = R3 = 2 0 kOhm ± 1 percent, 0 25 W

FIGURE 26 Output voltage swing test circuit for table XXIV.

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TABLE XXV. RMS-DC offset voltage test.

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C		Units	Fig
			Min	Max		
Z1	Offset voltage	B27	-1.5	1.5	mV	27
Z2	"	A14	"	"	"	"
Z3	Offset voltage	A28	-1.5	1.5	mV	27

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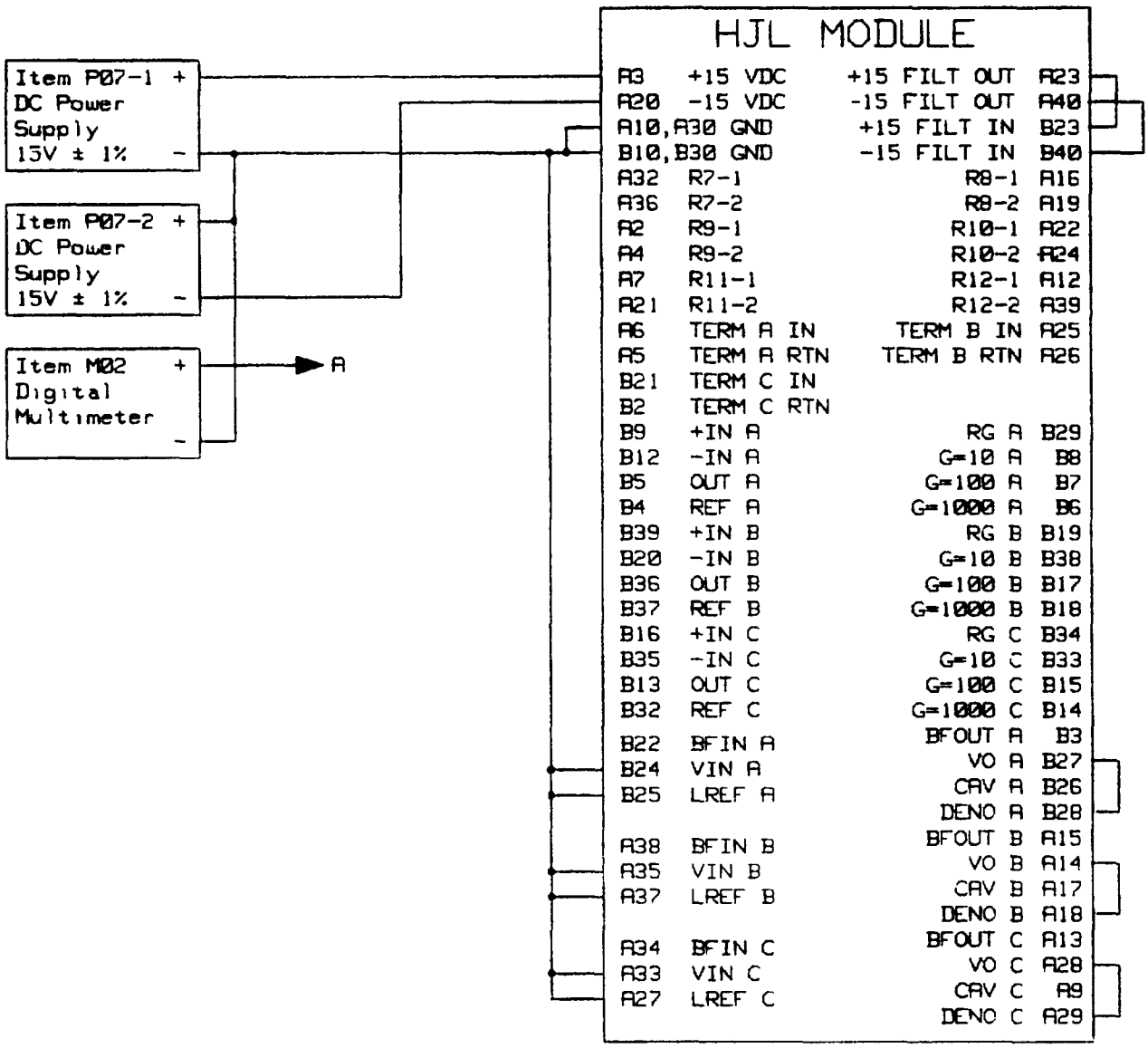


FIGURE 27 RMS-DC offset voltage test circuit for table XXV

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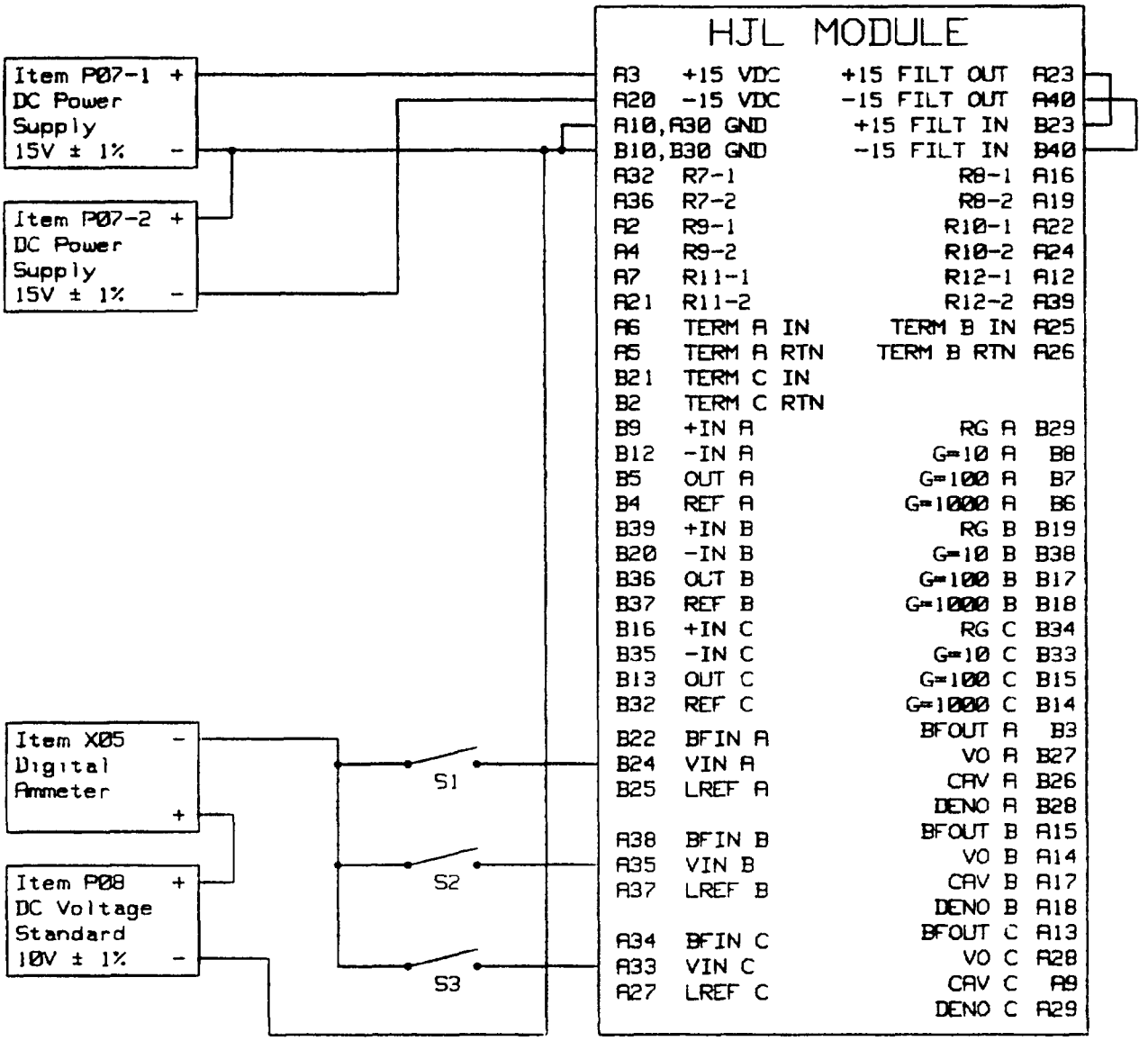
TABLE XXVI. RMS-DC input resistance test

Test cond	Parameter	Switch closed	Initial limits 25 ± 5°C		Units	Fig
			Min	Max		
AA1	Input resistance	S1	6.4	9.6	kOhm	28
AA2	|	S2	|	|	|	|
AA3	Input resistance	S3	6.4	9.6	kOhm	28

NOTE

1. The input resistance and measured input current are related by R = 10/I where I is the measured current for each test condition.

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NOTE

1 S1 = S2 = S3 = SPST switch.

FIGURE 28 RMS-DC input resistance test circuit for table XXVI.

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TABLE XXVII. RMS-DC nonlinearity test

Test cond	Parameter	Item P08 (Vdc) ± 0.01 V	Connect A to pin	Measurement/ calculation	Initial limits 25 ± 5°C		Units	Notes	Fig
					Min	Max			
	RMS-DC converter A	0.01	B27	#1A				1-6	29
	RMS-DC converter B	"	A14	#1B				"	"
	RMS-DC converter C	0.01	A28	#1C				"	"
	RMS-DC converter A	0.20	B27	#2A				"	"
	RMS-DC converter B	"	A14	#2B				"	"
	RMS-DC converter C	0.20	A28	#2C				"	"
	RMS-DC converter A	0.40	B27	#3A				"	"
	RMS-DC converter B	"	A14	#3B				"	"
	RMS-DC converter C	0.40	A28	#3C				"	"
	RMS-DC converter A	0.60	B27	#4A				"	"
	RMS-DC converter B	"	A14	#4B				"	"
	RMS-DC converter C	0.60	A28	#4C				"	"
	RMS-DC converter A	0 80	B27	#5A				"	"
	RMS-DC converter B	"	A14	#5B				"	"
	RMS-DC converter C	0 80	A28	#5C				"	"
	RMS-DC converter A	1.00	B27	#6A				"	"
	RMS-DC converter B	"	A14	#6B				"	"
	RMS-DC converter C	1 00	A28	#6C				"	"
	RMS-DC converter A	1.20	B27	#7A				"	"
	RMS-DC converter B	"	A14	#7B				"	"
	RMS-DC converter C	1.20	A28	#7C				"	"
	RMS-DC converter A	1.40	B27	#8A				"	"
	RMS-DC converter B	"	A14	#8B				"	"
	RMS-DC converter C	1.40	A28	#8C				"	"
	RMS-DC converter A	1.60	B27	#9A				"	"
	RMS-DC converter B	"	A14	#9B				"	"
	RMS-DC converter C	1 60	A28	#9C				1-6	29

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TABLE XXVII. RMS-DC nonlinearity test - Continued

Test cond	Parameter	Item P08 (Vdc) ± 0 01 V	Connect A to pin	Measurement/ calculation	Initial limits 25 ± 5°C		Units	Notes	Fig
					Min	Max			
	RMS-DC converter A	1 80	B27	#10A				1-6	29
	RMS-DC converter B	|	A14	#10B				|	|
	RMS-DC converter C	1.80	A28	#10C					
	RMS-DC converter A	2 00	B27	#11A					
	RMS-DC converter B	|	A14	#11B					
	RMS-DC converter C	2.00	A28	#11C				1-6	29
BB1	Converter A 2V range			See notes 1-6	-0 1	0.1	percent	1-6	29
BB2	Converter B 2V range			|	|	|	|	|	|
BB3	Converter C 2V range			See notes 1-6	-0 1	0.1	percent	1-6	29
	RMS-DC converter A	0 01	B27	#1A				1-6	29
	RMS-DC converter B	|	A14	#1B				|	|
	RMS-DC converter C	0.01	A28	#1C					
	RMS-DC converter A	0.50	B27	#2A					
	RMS-DC converter B	|	A14	#2B					
	RMS-DC converter C	0.50	A28	#2C					
	RMS-DC converter A	1.00	B27	#3A					
	RMS-DC converter B	|	A14	#3B					
	RMS-DC converter C	1.00	A28	#3C					
	RMS-DC converter A	1 50	B27	#4A					
	RMS-DC converter B	|	A14	#4B					
	RMS-DC converter C	1.50	A28	#4C					
	RMS-DC converter A	2.00	B27	#5A					
	RMS-DC converter B	|	A14	#5B					
	RMS-DC converter C	2.00	A28	#5C					
	RMS-DC converter A	2.50	B27	#6A					
	RMS-DC converter B	|	A14	#6B					
	RMS-DC converter C	2.50	A28	#6C				1-6	29

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TABLE XXVII RMS-DC nonlinearity test - Continued.

Test cond	Parameter	Item P08 (Vdc) ± 0.01 V	Connect A to pin	Measurement/ calculation	Initial limits 25 ± 5°C		Units	Notes	Fig
					Min	Max			
	RMS-DC converter A	3.00	B27	#7A				1-6	29
	RMS-DC converter B	|	A14	#7B				|	|
	RMS-DC converter C	3.00	A28	#7C				|	|
	RMS-DC converter A	3.50	B27	#8A				|	|
	RMS-DC converter B	|	A14	#8B				|	|
	RMS-DC converter C	3.50	A28	#8C				|	|
	RMS-DC converter A	4.00	B27	#9A				|	|
	RMS-DC converter B	|	A14	#9B				|	|
	RMS-DC converter C	4.00	A28	#9C				|	|
	RMS-DC converter A	4.50	B27	#10A				|	|
	RMS-DC converter B	|	A14	#10B				|	|
	RMS-DC converter C	4.50	A28	#10C				|	|
	RMS-DC converter A	5.00	B27	#11A				|	|
	RMS-DC converter B	|	A14	#11B				|	|
	RMS-DC converter C	5.00	A28	#11C				|	|
	RMS-DC converter A	5.50	B27	#12A				|	|
	RMS-DC converter B	|	A14	#12B				|	|
	RMS-DC converter C	5.50	A28	#12C				|	|
	RMS-DC converter A	6.00	B27	#13A				|	|
	RMS-DC converter B	|	A14	#13B				|	|
	RMS-DC converter C	6.00	A28	#13C				|	|
	RMS-DC converter A	6.50	B27	#14A				|	|
	RMS-DC converter B	|	A14	#14B				|	|
	RMS-DC converter C	6.50	A28	#14C				|	|
	RMS-DC converter A	7.00	B27	#15A				|	|
	RMS-DC converter B	|	A14	#15B				|	|
	RMS-DC converter C	7.00	A28	#15C				1-6	29

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TABLE XXVII. RMS-DC nonlinearity test - Continued.

Test cond	Parameter	Item P08 (Vdc) ± 0.01 V	Connect A to pin	Measurement/ calculation	Initial limits 25 ± 5°C		Units	Notes	Fig
					Min	Max			
BB4	Converter A 7V range			See notes 1-6	-0.1	0.1	percent	1-6	29
BB5	Converter B 7V range			⏐	⏐	⏐	⏐	⏐	⏐
BB6	Converter C 7V range			See notes 1-6	-0.1	0.1	percent	1-6	29

NOTES

1 The voltage reference shall be varied between 0 and 2 volts at 0.2 volt increments for the 2 volt range and between 0 and 7 volts at 0.5 volt increments for the 7 volt range. At each referenced voltage Vr(i), the output voltage Vo(i) measured with the multimeter shall be recorded

2 Nonlinearity error is defined in reference to the best line using the least squares method. The best line is: $Vo(i) = mVr(i) + b$.

3 Computation of the best line using the least squares method is defined by:

$$m = \frac{n\sum Vr(i)\,Vo(i) - \sum Vo(i)\sum Vr(i)}{n\sum Vr(i)^2 - (\sum Vr(i))^2}$$

$$b = \overline{Vo(i)} - \overline{Vr(i)}$$

where n is the number of data points, n = 11 for 2 volt range and n = 15 for 7 volt range.

4. All superscripted lines denote averages, hence all measurements must be made prior to performing computations

5 Nonlinearity error defined in reference to the best line using the least squares method is:

$$\text{Nonlinearity} = \frac{(m\,Vr(i) + b - Vo(i))\ \max}{FSR} * 100$$

where FSR is the full scale range, 2 volts or 7 volts.

6 Nonlinearity error shall be within the limits specified in table XXVII.

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TABLE XXVIII. RMS-DC PSS test.

Test cond	Parameter	Item P07-1 (Vdc) ± 1 percent	Item P07-2 (Vdc) ± 1 percent	Item P08 (Vdc) ± 0.01 V	Connect A to pin	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
							Min	Max		
		15.0	-15.0	0 0	B27	#1				29
		|	|	|	A14	#2				|
		|	|	0.0	A28	#3				|
		|	|	7.0	B27	#4				|
		|	|	|	A14	#5				|
		15 0	-15 0	7 0	A28	#6				29
CC1	PSS	13.5	-13 5	0 0	B27	CC1=|(Vo-#1)|/3 0	0	300	uV/V	29
CC2	|	|	|	|	A14	CC2=|(Vo-#2)|/3.0	|	|	|	|
CC3	|	|	|	0 0	A28	CC3=|(Vo-#3)|/3.0	|	|	|	|
CC4	|	|	|	7.0	B27	CC4=|(Vo-#4)|/3.0	|	|	|	|
CC5	|	|	|	|	A14	CC5=|(Vo-#5)|/3 0	|	|	|	|
CC6	|	13 5	-13.5	7 0	A28	CC6=|(Vo-#6)|/3 0	|	|	|	|
CC7	|	16 5	-16 5	0 0	B27	CC7=|(Vo-#1)|/3 0	|	|	|	|
CC8	|	|	|	|	A14	CC8=|(Vo-#2)|/3.0	|	|	|	|
CC9	|	|	|	0.0	A28	CC9=|(Vo-#3)|/3.0	|	|	|	|
CC10	|	|	|	7.0	B27	CC10=|(Vo-#4)|/3.0	|	|	|	|
CC11	|	|	|	|	A14	CC11=|(Vo-#5)|/3.0	|	|	|	|
CC12	PSS	16 5	-16.5	7 0	A28	CC12=|(Vo-#6)|/3.0	0	300	uV/V	29

NOTES:

1. Vo in the above table represents the measurement taken for each step.

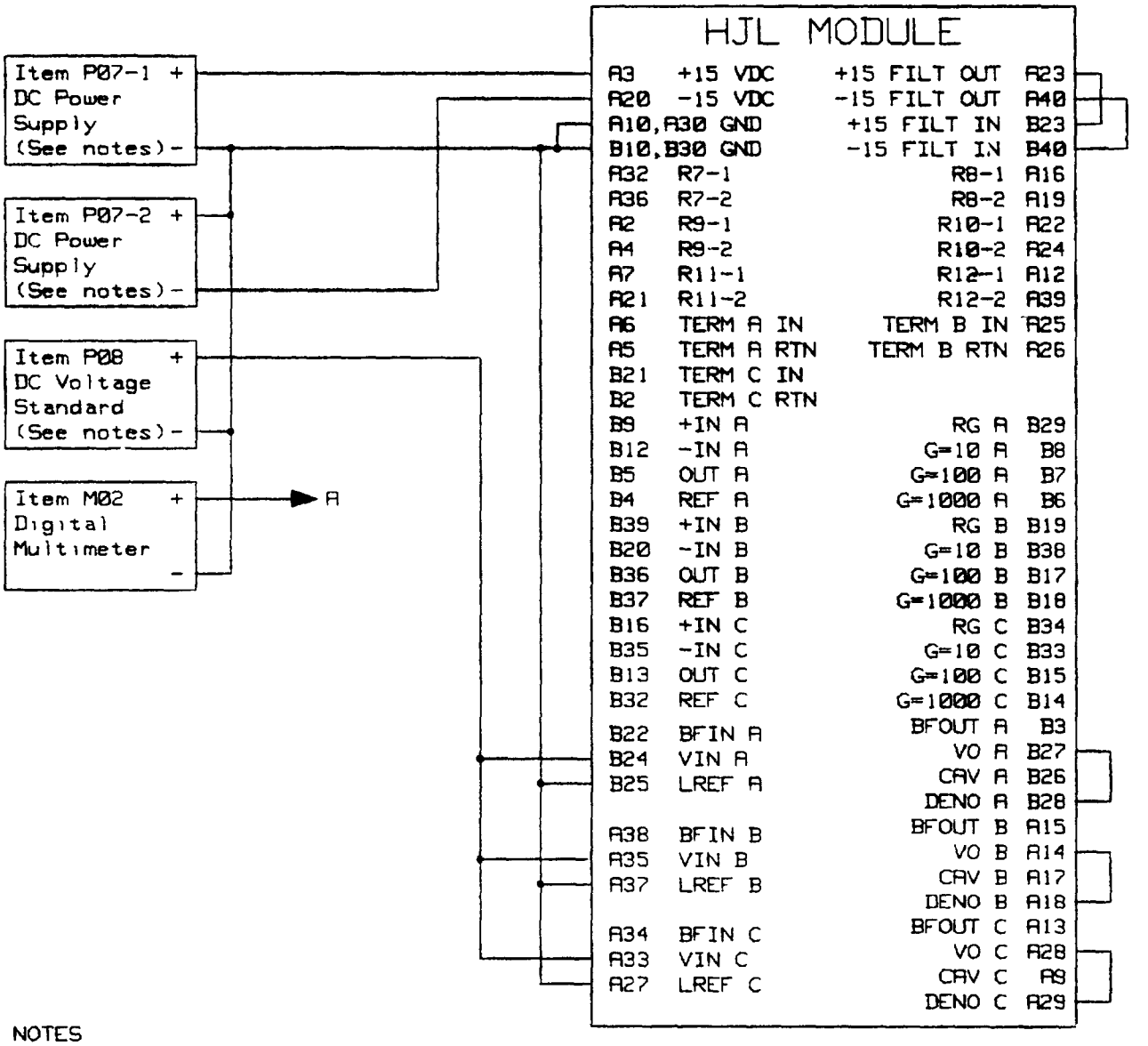
2. Abbreviations used are defined in 6.1.

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TABLE XXIX. RMS-DC reversal error test.

Test cond	Parameter	Connect A to pin	Item P08 (Vdc) ± 0.01 V	Measurement/ computation	Initial limits 25 ± 5°C		Units	Fig
					Min	Max		
		B27	5.0	#1				29
		A14	|	#2				|
		A28	5 0	#3				|
		B27	-5 0	#4				|
		A14	|	#5				|
		A28	-5.0	#6				29
DD1	Reversal error RMS-DC A			DD1=100*(#1-#4)/5	-0.25	0.25	percent	29
DD2	Reversal error RMS-DC B			DD2=100*(#2-#5)/5	|	|	|	|
DD3	Reversal error RMS-DC C			DD3=100*(#3-#6)/5	-0.25	0.25	percent	29

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NOTES

1 For test conditions BB1-BB6 and DD1-DD3, Item P07-1 = 15 V ± 1 percent and Item P07-2 = -15 V ± 1 percent

2 Adjust Item P08 as specified in tables XXVII, XXVIII, and XXIX.

FIGURE 29 RMS-DC nonlinearity, PSS and reversal error test circuit for tables XXVII, XXVIII, and XXIX

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TABLE XXX. RMS-DC total error test.

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C		Units	Fig
			Min	Max		
EE1	Total error	B27	6.964	7.036	V	30
EE2	|	A14	|	|	|	|
EE3	Total error	A28	6.964	7.036	V	30

NOTE:

1 The limits above result from ± 1 mV ± 0.5 percent of reading.

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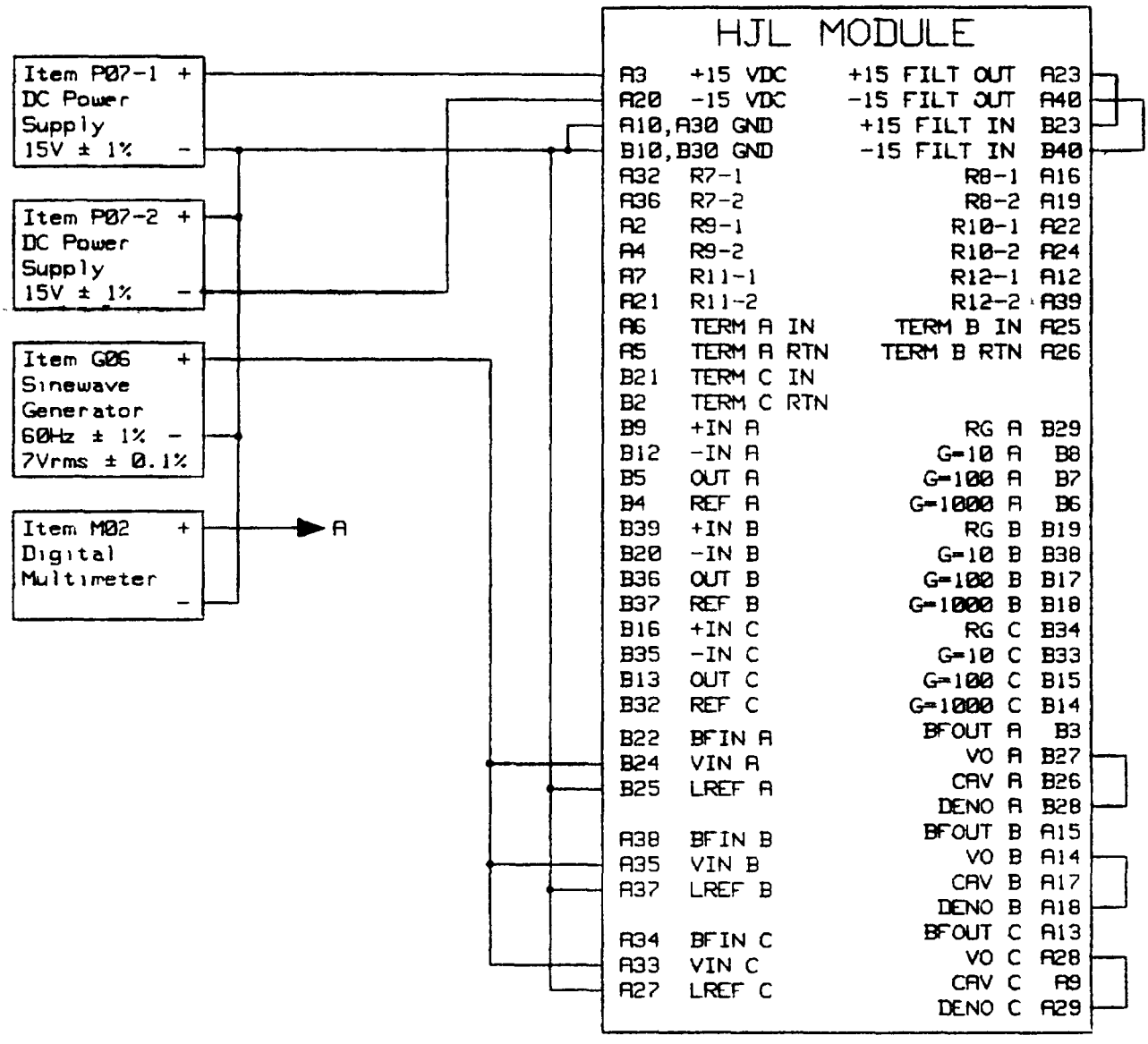


FIGURE 30 RMS-DC total error test circuit for table XXX.

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TABLE XXXI. Crest factor error test.

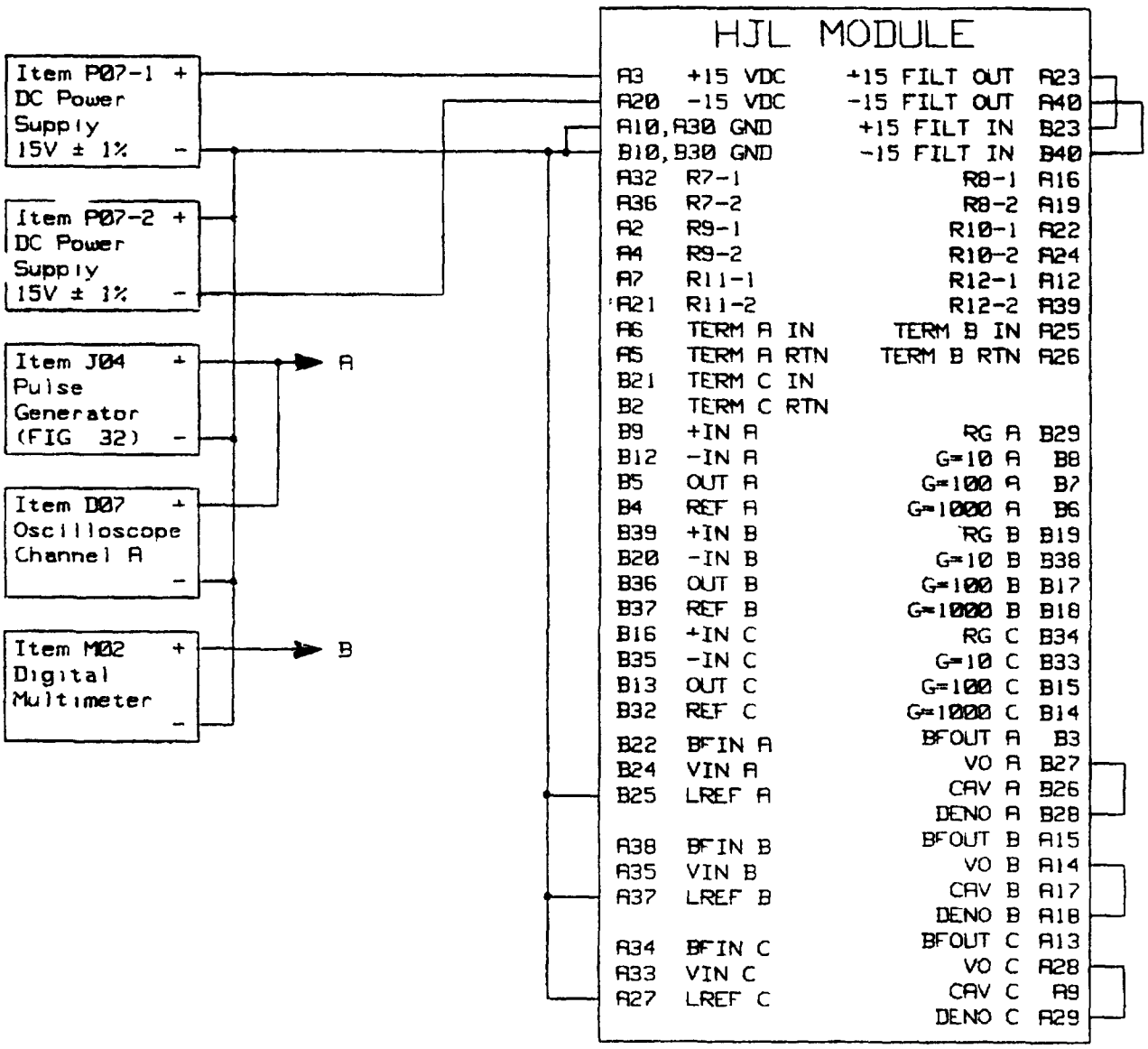
Test cond	Parameter	Connect		Computation	Initial limits 25 ± 5°C		Units	Figs
		A to pin	B to pin		Min	Max		
FF1	Crest factor error	B24	B27	FF1=(1-Vo)*100	-5.0	5.0	percent	31,32
FF2	|	A35	A14	FF2=(1-Vo)*100	|	|	|	|
FF3	Crest factor error	A33	A28	FF3=(1-Vo)*100	-5.0	5.0	percent	31,32

NOTES:

1. The input waveform setup is shown on figure 32.

2. In the above equations Vo represents the measurement taken for that step.

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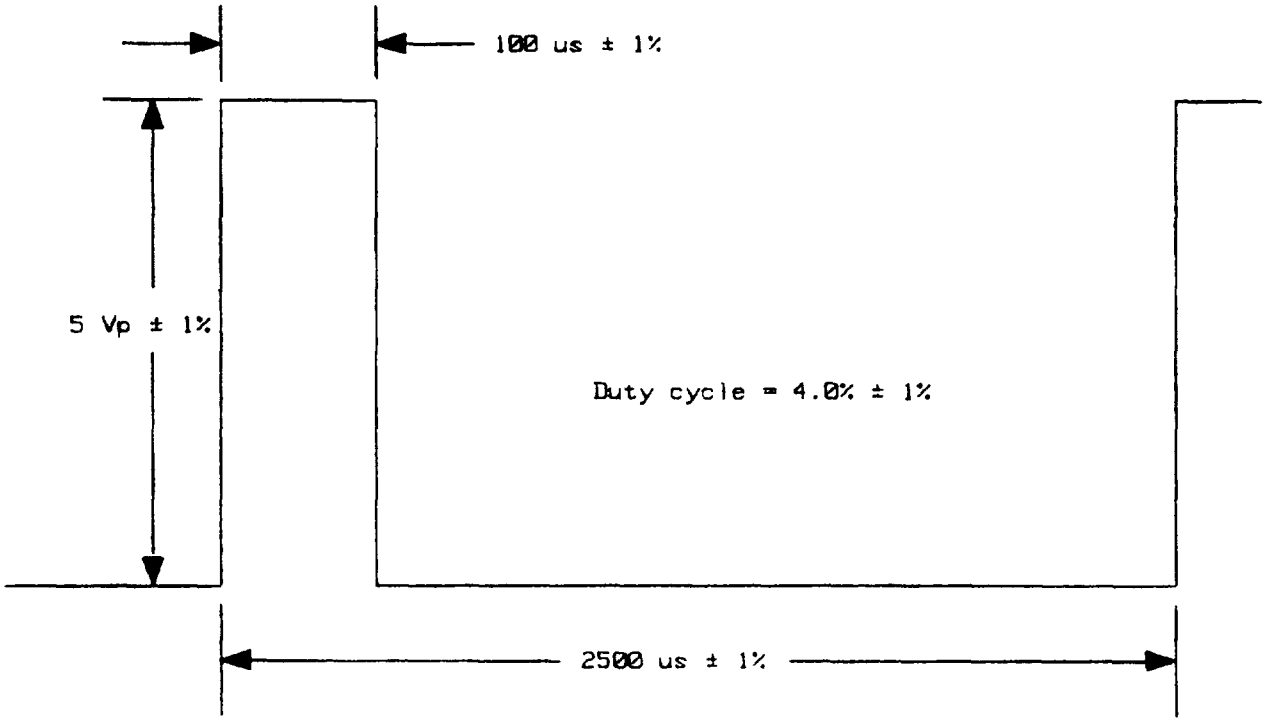


NOTE:

1. Scope probe = Y02.

FIGURE 31 Crest factor error test circuit for table XXXI

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Crest factor = 5

FIGURE 32 Input waveform for crest factor test for figure 31

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TABLE XXXII. Ripple voltage test.

Test cond	Parameter	Connect A to pin	Initial limits 25 ± 5°C		Units	Fig
			Min	Max		
GG1	Ripple voltage	B27	54	115	mVrms	33
GG2	"	A14	"	"	"	"
GG3	Ripple voltage	A28	54	115	mVrms	33

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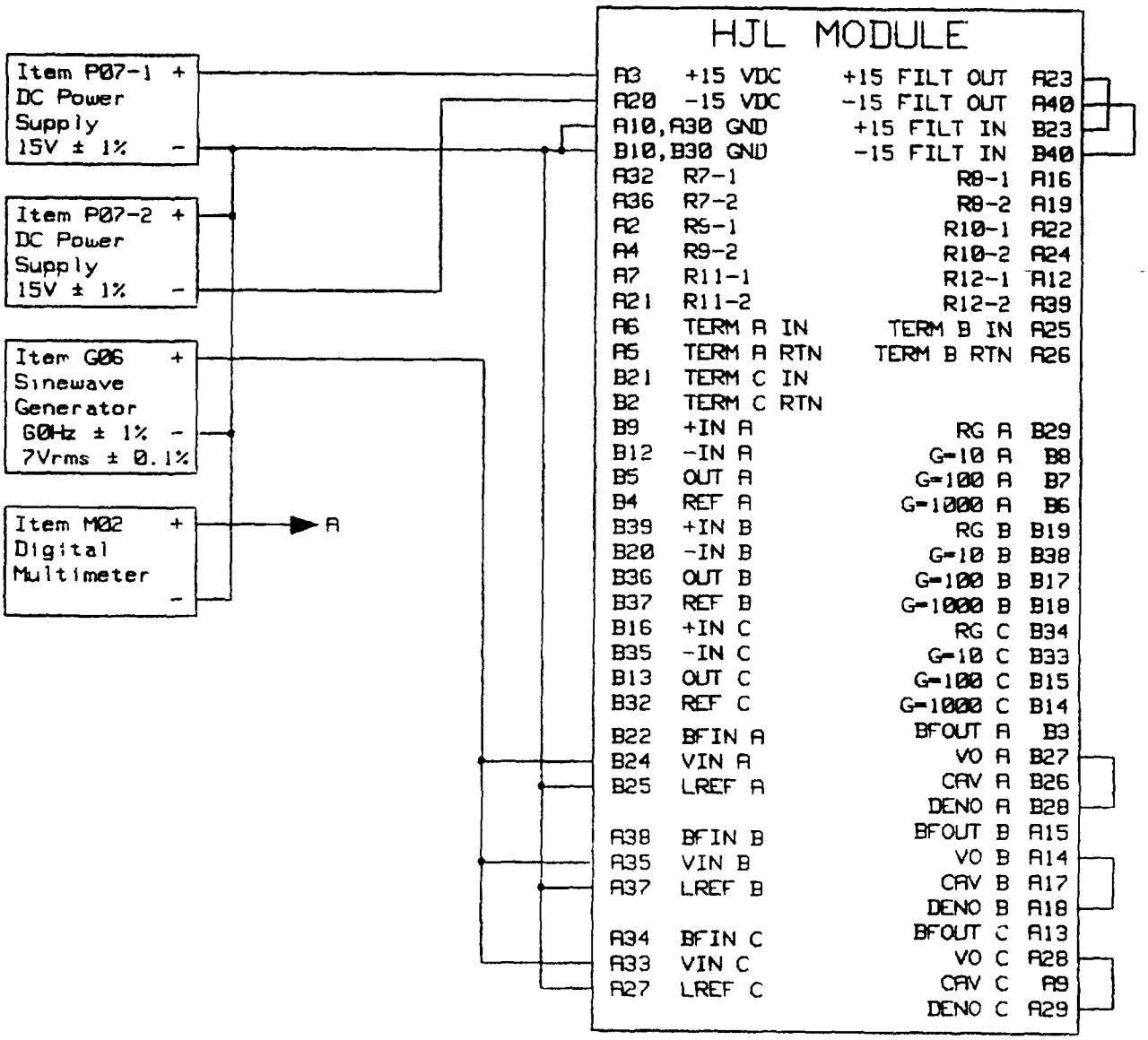


FIGURE 33 Ripple voltage test circuit for table XXXII

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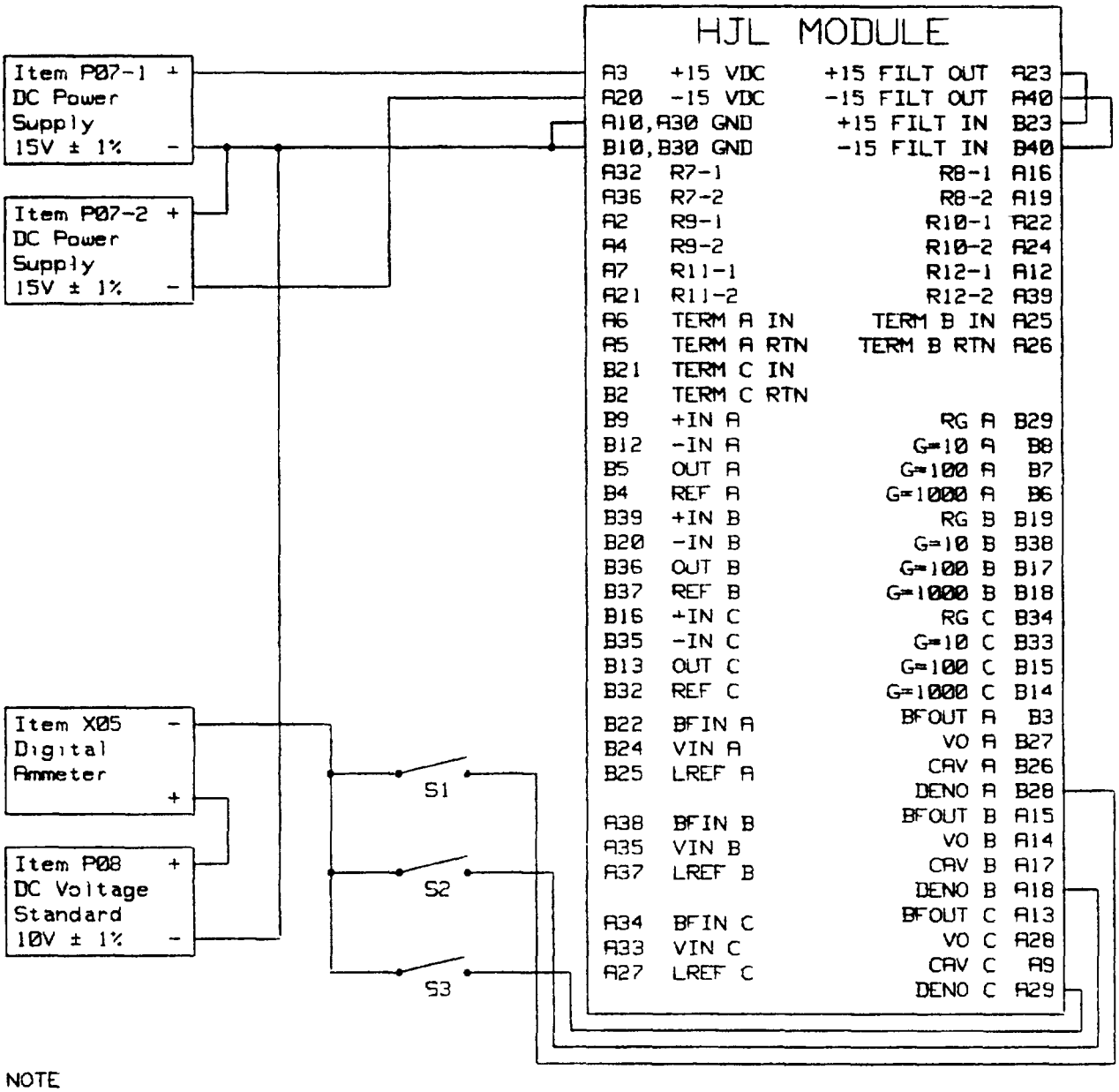
TABLE XXXIII. RMS-DC input resistance test

Test cond	Parameter	Switch closed	Initial limits 25 ± 5°C		Units	Fig
			Min	Max		
HH1	Input resistance	S1	20	30	kOhm	34
HH2	|	S2	|	|	|	|
HH3	Input resistance	S3	20	30	kOhm	34

NOTE:

1. The input resistance and measured input current are related by R = 10/I where I is the measured current for each test condition.

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NOTE

1 S1 = S2 = S3 = SPST switch

FIGURE 34 RMS-DC input resistance test circuit for table XXXIII

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TABLE XXXIV. Frequency response test.

Test cond	Parameter	Item G06		Connect		Initial limits 25 ± 5°C		Units	Fig
		(kHz) ± 1 percent	(mVrms) ± 1 percent	A to pin	B to pin	Min	Max		
KK1	Additional error	10	20	B24	B27	-4.0	4.0	percent	35
KK2				A35	A14				
KK3		10	20	A33	A28				
KK4		60	200	B24	B27				
KK5				A35	A14				
KK6		60	200	A33	A28				
KK7		200	2000	B24	B27				
KK8				A35	A14				
KK9	Additional error	200	2000	A33	A28	-4.0	4.0	percent	35

NOTES

1. Vo is the value measured in each step.

2. The additional error for each step is calculated as: percent error = 100*(Vin-Vo)/Vin.

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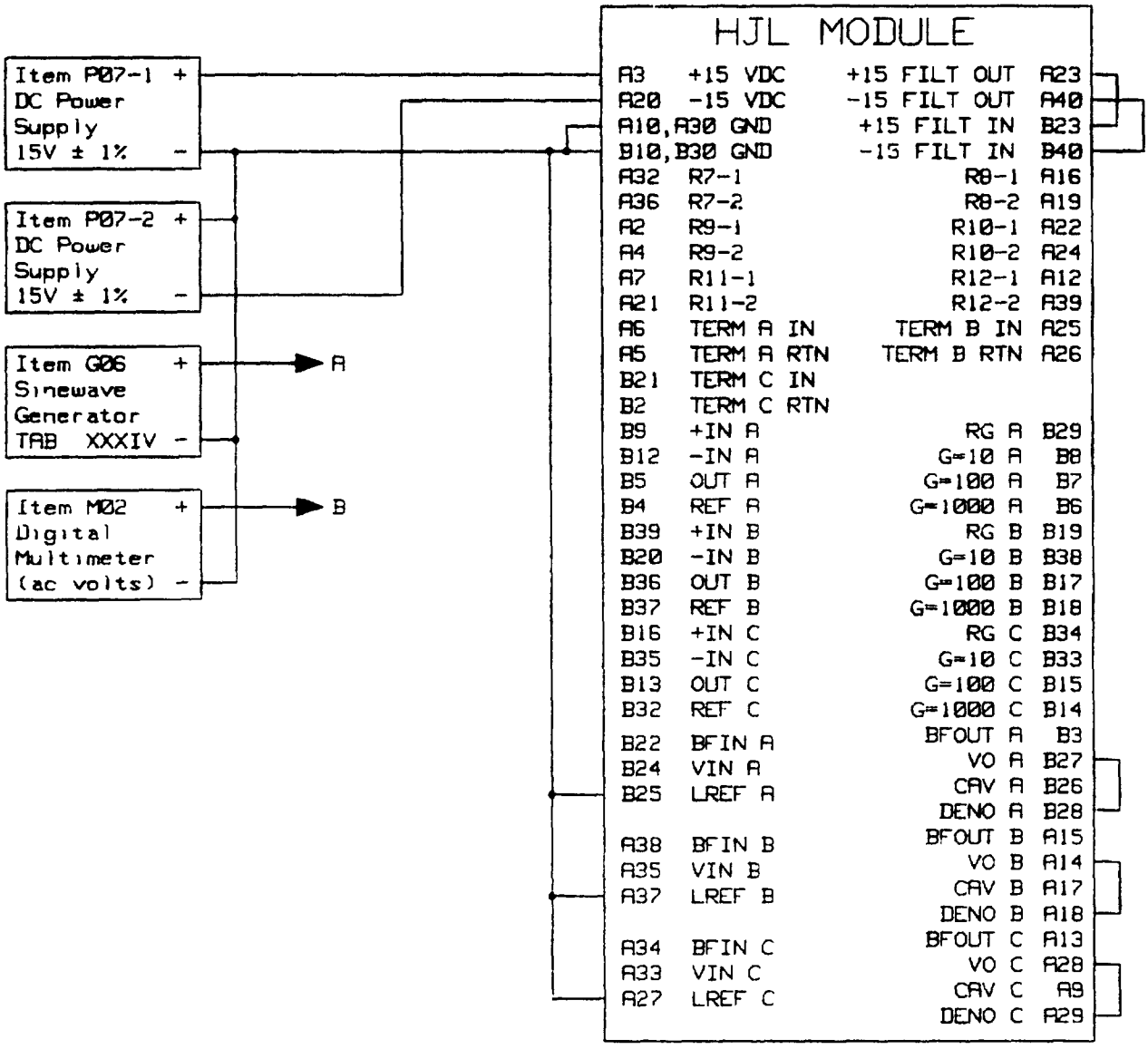


FIGURE 35 Frequency response test circuit for table XXXIV

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-W-28787/370	1990 October 26

3 DOCUMENT TITLE
Module, Standard Electronic Network, Special, Keycode HJL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-1528	88-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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